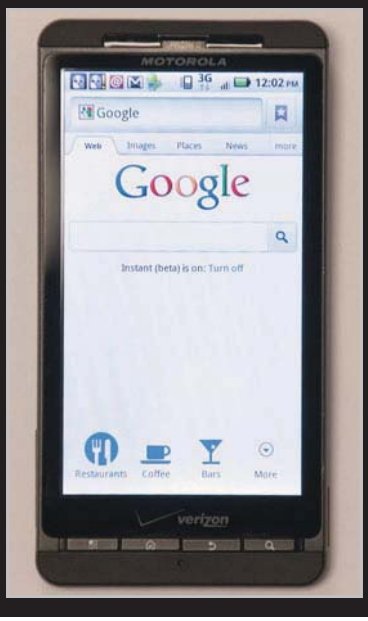
14

FORECASTING DEMAND



KEVIN DIETSCH/UPI/Newscom

A Motorola Droid phone displayed Google's home-page in Washington, D.C. on August 15, 2011. Google Inc. bought the phone manufacturer Motorola Mobility for $12.5 billion. Motorola considerably improved its demand forecasting process, with payoffs in how it managed its supply chain.

Motorola Mobility

Motorola Mobility makes mobile phone handsets, smartphones, tablets, and cable set-top box assets. In the early 2000s, Motorola's leadership and market share were eroding. Motorola realized that it must transform its supply chain, and embarked on a major initiative to tighten communications and collaboration along its supply chain. It put collaborative planning, forecasting, and replenishment (*CPFR*) into action in 2002. The payoff has been significant.

Motorola sells over 120 handset models globally. Forecasting how many of which models to make and sell is difficult, and accurate replenishment of retailers' shelves is critical. If a customer's favorite handset is not in stock, there is a real risk that Motorola loses that customer for life, and not just for the next service contract. Approximately one half of all stockouts result in lost sales. To make matters worse, a phone model can have multiple SKUs, life cycles average little more than a year, and new product introductions are rapid.

Prior to adopting CPFR, Motorola Mobile's sales were highly variable and were not synchronized with customer demand. Motorola had visibility only for its shipments to retailers' distribution centers, but not for shipments from the retailers' distribution centers to the stores. Knowing what retailers are selling is much more valuable information in forecasting future demand than knowing what retailers are buying. Without this information, forecast errors were very high, resulting in excessive stockouts. CPFR enabled Motorola to collaborate with

its retailers' distribution centers' customers and increase its ability to forecast effectively. Motorola launched an organization-wide shift to customer-focused operations teams. They shared with their retailers their real-time data and plans, including forecasts, inventories, sales to retailers' shelves, promotions, product plans, and exceptions. Traditionally, suppliers and buyers in most supply chains prepare independent demand forecasts.

Before CPFR, the retailers' forecasts were developed at the end of the second week of each month while Motorola's assembled its sales and operations plan earlier in the second week. Motorola convinced the retailer to move up its planning cycle by just two or three days, which eliminated a seven-week forecast lag resulting from the forecast not being incorporated until the next month's planning cycle. Now, the retailer loads its forecasts for the next month on Monday. On Tuesday, Motorola loads its forecast. During the weekly call on Wednesday, the two teams jointly resolve discrepancies line-by-line. The inclusion of a forecasting analyst means they can immediately resolve issues arising from the discrepancies.

The real key to a successful implementation of CPFR is the forging of a cultural alliance that involves peer-to peer relations and cross-functional teams. Prior to CPFR, retailers sometimes gave Motorola "C," "D," and "F" rating on metrics such as on-time delivery, ease of doing business, and stockouts. After CPFR, they give Motorola "A" ratings. Motorola's CPFR initiative reduced forecast error to a fraction of its previous level, allowed quick reductions in safety stock, cut transportation costs in half because of fewer less-than-truckload shipments, and cut stockouts to less than a third of previous levels. Such success is one reason Google paid big ($12.5 billion) to buy Motorola's cellphone business in August 2011.

Source: Jerold P. Cederlund, Rajiv Kohli, Susan A. Sherer, and Yuliang Yao, "How Motorola Put CPFR into Action," *Supply Chain Management Review* (October 2007), pp. 28–35; Sharyn Leaver, Patrick Connaughton, and Elisse Gaynor, "Case Study: Motorola's Quest for Supply Chain Excellence," *Forrester Research, Inc.* (October, 2006), pp. 1–12, **www.motorola.com**, April 29, 2011; Amir Efrati and Spencer A. Ante, "Google's $12.5 Billion Gamble," *The Wall Street Journal*, August 12, 2011.

LEARNING GOALS *After reading this chapter, you should be able to:*

1. Identify the five basic patterns of most demand time series.
2. Identify the various measures of forecast errors.
3. Use regression to make forecasts with one or more independent variables.
4. Make forecasts using the most common approaches for time-series analysis.
5. Make forecasts using trend projection with regression.
6. Describe a typical forecasting process used by businesses.
7. Explain collaborative planning, forecasting, and replenishment (CPFR).

forecast
A prediction of future events used for planning purposes.

Balancing supply and demand begins with making accurate forecasts, and then reconciling them across the supply chain as shown by Motorola Mobility. A **forecast** is a prediction of future events used for planning purposes. Planning, on the other hand, is the process of making management decisions on how to deploy resources to best respond to the demand forecasts. Forecasting methods may be based on mathematical models that use available historical data, or on qualitative methods that draw on managerial experience and judgments, or on a combination of both.

In this chapter, our focus is on demand forecasts. We begin with different types of demand patterns. We examine forecasting methods in three basic categories: (1) judgment, (2) causal, and (3) time-series methods. Forecast errors are defined, providing important clues for making better forecasts. We next consider the forecasting techniques themselves, and then how they can be combined to bring together insights from several sources. We conclude with overall processes for making forecasts and designing the forecasting system.

Forecasts are useful for both managing processes and managing supply chains. At the supply chain level, a firm needs forecasts to coordinate with its customers and suppliers. At the process level, output forecasts are needed to design the various processes throughout the organization, including identifying and dealing with in-house bottlenecks.

Competing with Operations
Using Operations to Compete
Managing Effective Projects

Designing and Managing Processes
Developing a Process Strategy
Analyzing Processes
Managing Quality
Planning Capacity
Managing Process Constraints
Designing Lean Systems

Designing and Managing Supply Chains
Managing Inventories
Designing Effective Supply Chains
Locating Facilities
Integrating the Supply Chain
Managing Sustainable Supply Chains
Forecasting Demand
Planning and Scheduling Operations
Planning Sufficient Resources

Forecasting across the Organization

The organization-wide forecasting process cuts across functional areas. Forecasting overall demand typically originates with marketing, but internal customers throughout the organization depend on forecasts to formulate and execute their plans as well. Forecasts are critical inputs to business plans, annual plans, and budgets. Finance needs forecasts to project cash flows and capital requirements. Human resources uses forecasts to anticipate hiring and training needs. Marketing is an important source for sales forecast information because it is closest to external customers. Operations and supply chain managers need forecasts to plan output levels, purchases of services and materials, workforce and output schedules, inventories, and long-term capacities.

Managers throughout the organization make forecasts on many variables other than future demand, such as competitor strategies, regulatory changes, technological changes, processing times, supplier lead times, and quality losses. Tools for making these forecasts are basically the same tools covered here for demand forecasting: judgment, opinions of knowledgeable people, averages of experience, regression, and time-series techniques. Using these tools, forecasting can be improved. Still, forecasts are rarely perfect. As Samuel Clemens (Mark Twain) said in *Following the Equator*, "Prophesy is a good line of business, but it is full of risks." Smart managers recognize this reality and find ways to update their plans when the inevitable forecast error or unexpected event occurs.

Demand Patterns

Forecasting customer demand is a difficult task because the demand for services and goods can vary greatly. For example, demand for lawn fertilizer predictably increases in the spring and summer months; however, the particular weekends when demand is heaviest may depend on uncontrollable factors such as the weather. Sometimes, patterns are more predictable. Thus, the peak hours of the day for a large bank's call center are from 9:00 A.M. to 12:00 P.M., and the peak day of the week is Monday. For its statement-rendering processes, the peak months are January, April, July, and October, which is when the quarterly statements are sent out. Forecasting demand in such situations requires uncovering the underlying patterns from available information. In this section, we discuss the basic patterns of demand.

The repeated observations of demand for a service or product in their order of occurrence form a pattern known as a **time series**. There are five basic patterns of most demand time series:

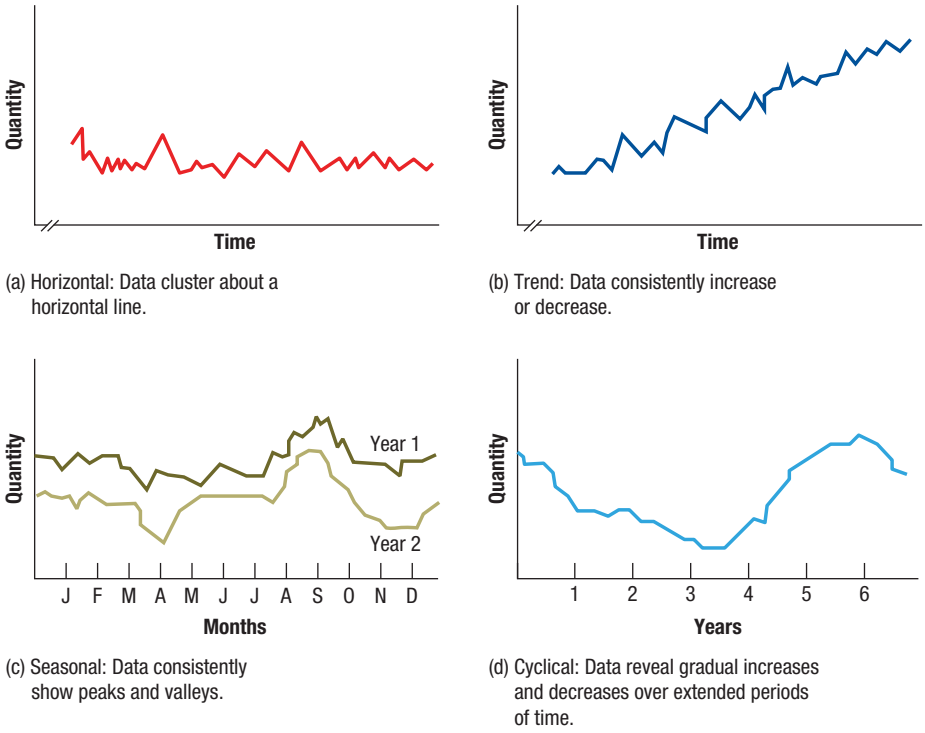
time series
The repeated observations of demand for a service or product in their order of occurrence.

1. *Horizontal.* The fluctuation of data around a constant mean.
2. *Trend.* The systematic increase or decrease in the mean of the series over time.
3. *Seasonal.* A repeatable pattern of increases or decreases in demand, depending on the time of day, week, month, or season.
4. *Cyclical.* The less predictable gradual increases or decreases in demand over longer periods of time (years or decades).
5. *Random.* The unforecastable variation in demand.

Cyclical patterns arise from two influences. The first is the business cycle, which includes factors that cause the economy to go from recession to expansion over a number of years. The other influence is the service or product life cycle, which reflects the stages of demand from development through decline. Business cycle demand is difficult to predict because it is affected by national or international events.

The four patterns of demand—horizontal, trend, seasonal, and cyclical—combine in varying degrees to define the underlying time pattern of demand for a service or product. The fifth pattern, random variation, results from chance causes and thus, cannot be predicted. Random variation is an aspect of demand that makes every forecast ultimately inaccurate. Figure 14.1 shows the first four patterns of a demand time series, all of which contain random variations.

FIGURE 14.1 ▶
Patterns of Demand



(a) Horizontal: Data cluster about a horizontal line.

(b) Trend: Data consistently increase or decrease.

(c) Seasonal: Data consistently show peaks and valleys.

(d) Cyclical: Data reveal gradual increases and decreases over extended periods of time.

Key Decisions on Making Forecasts

Before using forecasting techniques, a manager must make two decisions: (1) what to forecast, and (2) what type of forecasting technique to select for different items.

Deciding What to Forecast

Although some sort of demand estimate is needed for the individual services or goods produced by a company, forecasting total demand for groups or clusters and then deriving individual service or product forecasts may be easiest. Also, selecting the correct unit of measurement (e.g., service or product units or machine-hours) for forecasting may be as important as choosing the best method.

Level of Aggregation Few companies err by more than 5 percent when forecasting the annual total demand for all their services or products. However, errors in forecasts for individual items and shorter time periods may be much higher. Recognizing this reality, many companies use a two-tier forecasting system. They first cluster (or "roll up") several similar services or products in a process called **aggregation**, making forecasts for families of services or goods that have similar demand requirements and common processing, labor, and materials requirements. Next, they derive forecasts for individual items, which are sometimes called stock-keeping units. A *stock-keeping unit (SKU)* is an individual item or product that has an identifying code and is held in inventory somewhere along the supply chain, such as in a distribution center.

aggregation
The act of clustering several similar services or products so that forecasts and plans can be made for whole families.

Units of Measurement Rather than using dollars as the initial unit of measurement, forecasts often begin with service or product units, such as SKUs, express packages to deliver, or customers needing maintenance service or repairs for their cars. Forecasted units can then be translated to dollars by multiplying them by the unit price. If accurately forecasting demand for a service or product is not possible in terms of number of units, forecast the standard labor or machine-hours required of each of the critical resources.

Choosing the Type of Forecasting Technique

judgment methods
A forecasting method that translates the opinions of managers, expert opinions, consumer surveys, and salesforce estimates into quantitative estimates.

Forecasting systems offer a variety of techniques, and no one of them is best for all items and situations. The forecaster's objective is to develop a useful forecast from the information at hand with the technique that is appropriate for the different patterns of demand. Two general types of forecasting techniques are used: judgment methods and quantitative methods. **Judgment methods** translate the opinions of managers, expert opinions, consumer surveys, and salesforce estimates

into quantitative estimates. Quantitative methods include causal methods, time-series analysis, and trend projection with regression. **Causal methods** use historical data on independent variables, such as promotional campaigns, economic conditions, and competitors' actions, to predict demand. **Time-series analysis** is a statistical approach that relies heavily on historical demand data to project the future size of demand and recognizes trends and seasonal patterns. **Trend projection using regression** is a hybrid between a time-series technique and the causal method.

causal methods

A quantitative forecasting method that uses historical data on independent variables, such as promotional campaigns, economic conditions, and competitors' actions, to predict demand.

time-series analysis

A statistical approach that relies heavily on historical demand data to project the future size of demand and recognizes trends and seasonal patterns.

trend projection with regression

A forecasting model that is a hybrid between a time-series technique and the causal method.

Forecast Error

For any forecasting technique, it is important to measure the accuracy of its forecasts. Forecasts almost always contain errors. Random error results from unpredictable factors that cause the forecast to deviate from the actual demand. Forecasting analysts try to minimize forecast errors by selecting appropriate forecasting models, but eliminating all forms of errors is impossible.

Forecast error for a given period t is simply the difference found by subtracting the forecast from actual demand, or

$$E_t = D_t - F_t$$

where

$$E_t = \text{forecast error for period } t$$
$$D_t = \text{actual demand for period } t$$
$$F_t = \text{forecast for period } t$$

This equation (notice the alphabetical order with D_t coming before F_t) is the starting point for creating several measures of forecast error that cover longer periods of time. Figure 14.2 shows the output from the *Error Analysis* routine in Forecasting's dropdown menu of POM for Windows. Part (a) gives a big picture view of how well the forecast has been tracking the actual demand. Part (b) shows the detailed calculations needed to obtain the summary error terms. Finally, Part (c) gives the summary error measures summarized across all 10 time periods, as derived from Part (b).

forecast error

The difference found by subtracting the forecast from actual demand for a given period.

The **cumulative sum of forecast errors (CFE)** measures the total forecast error:

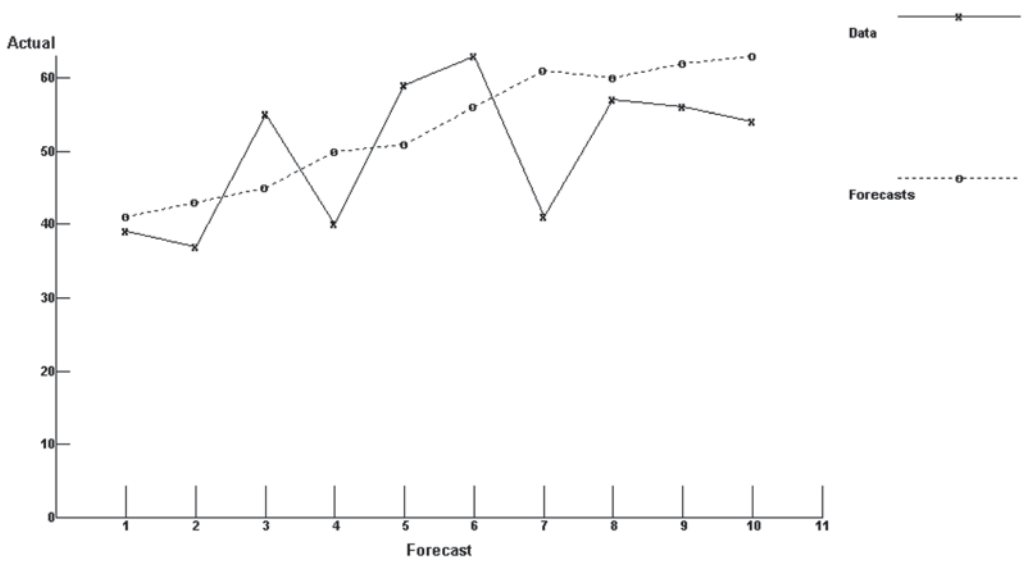
$$\text{CFE} = \sum E_t$$

CFE is a cumulative sum. Figure 14.3(b) shows that it is the sum of the errors for all 10 periods. For any given period, it would be the sum of errors up through that period. For example, it would be −8 (or −2 −6) for period 2. CFE is also called the *bias error* and results from consistent mistakes—the forecast is always too high or too low. This type of error typically causes the greatest disruption to planning efforts. For example, if a forecast is consistently lower than actual demand, the value of CFE will gradually get larger and larger. This increasingly large error indicates some systematic deficiency in the forecasting approach. The average forecast error, sometimes called the *mean bias*, is simply

cumulative sum of forecast errors (CFE)

A measurement of the total forecast error that assesses the bias in a forecast.

$$\bar{E} = \frac{\text{CFE}}{n}$$



◀ **FIGURE 14.2(a)**
Graph of Actual and Forecast Demand Using *Error Analysis* of Forecasting in POM for Windows

▶ **FIGURE 14.2(b)**
Detailed Calculations of Forecast Errors

		Forecast	Error	|Error|	Error^2	|Pct Error|
Past period 1	39	41	-2	2	4	5.128%
Past period 2	37	43	-6	6	36	16.216%
Past period 3	55	45	10	10	100	18.182%
Past period 4	40	50	-10	10	100	25%
Past period 5	59	51	8	8	64	13.559%
Past period 6	63	56	7	7	49	11.111%
Past period 7	41	61	-20	20	400	48.78%
Past period 8	57	60	-3	3	9	5.263%
Past period 9	56	62	-6	6	36	10.714%
Past period 10	54	63	-9	9	81	16.667%
TOTALS	501		-31	81	879	170.621%
AVERAGE	50.1		-3.1	8.1	87.9	17.062%
			(Bias)	(MAD)	(MSE)	(MAPE)
				Std dev	29.648	

mean squared error (MSE)
A measurement of the dispersion of forecast errors.

standard deviation (σ)
A measurement of the dispersion of forecast errors.

mean absolute deviation (MAD)
A measurement of the dispersion of forecast errors.

The **mean squared error (MSE)**, **standard deviation of the errors** (σ), and **mean absolute deviation (MAD)** measure the dispersion of forecast errors attributed to trend, seasonal, cyclical, or random effects:

$$\text{MSE} = \frac{\Sigma E_t^2}{n}$$

$$\sigma = \sqrt{\frac{\Sigma (E_t - \bar{E})^2}{n-1}}$$

$$\text{MAD} = \frac{\Sigma |E_t|}{n}$$

Figure 14.2(b) shows the squared error in period 1 is 4, and MSE is 87.9 for the whole sample. The standard deviation of the errors is calculated using one of the functions available in Excel and is not shown in Figure 14.2(b). The absolute value of the error in period 2 is 6, and MAD is 8.1 across the whole sample.

The mathematical symbol | | is used to indicate the absolute value—that is, it tells you to disregard positive or negative signs. If MSE, σ, or MAD is small, the forecast is typically close to actual demand; by contrast, a large value indicates the possibility of large forecast errors. The measures do differ in the way they emphasize errors. Large errors get far more weight in MSE and σ because the errors are squared. MAD is a widely used measure of forecast error and is easily understood; it is merely the mean of the absolute forecast errors over a series of time periods, without regard to whether the error was an overestimate or an underestimate.

mean absolute percent error (MAPE)
A measurement that relates the forecast error to the level of demand and is useful for putting forecast performance in the proper perspective.

The **mean absolute percent error (MAPE)** relates the forecast error to the level of demand and is useful for putting forecast performance in the proper perspective:

$$\text{MAPE} = \frac{(\Sigma |E_t|/D_t)\,(100)}{n} \text{ (expressed as a percentage)}$$

▼ **FIGURE 14.2(c)**
Error Measures

Measure	Value
Error Measures	
CFE (Cumulative Forecast Error)	-31
MAD (Mean Absolute Deviation)	8.1
MSE (Mean Squared Error)	87.9
Standard Deviation of Errors	29.648
MAPE (Mean Absolute Percent	17.062%

For example, an absolute forecast error of 100 results in a larger percentage error when the demand is 200 units than when the demand is 10,000 units. MAPE is the best error measure to use when making comparisons between time series for different SKUs. Looking again at Figure 14.2(b), the percent error in period 2 is 16.22 percent, and MAPE, the average over all 10 periods, is 17.06 percent.

Finally, Figure 14.2(c) summarizes the key error terms across all 10 time periods. They are actually found in selected portions of Figure 14.2(b). For example, CFE is -31, which is in the error column of Figure 14.2(b) in the TOTALS row. MAD is 8.1, found in the |Error| column and AVERAGE row. Finally, | | is 17.06%, which is in the |Pct Error| column and AVERAGE row.

EXAMPLE 14.1 Calculating Forecast Error Measures

The following table shows the actual sales of upholstered chairs for a furniture manufacturer and the forecasts made for each of the last 8 months. Calculate CFE, MSE, σ, MAD, and MAPE for this product.

Month, t	Demand, D_t	Forecast, F_t	Error, E_t	Error, Squared, E_t^2	Absolute Error $\lvert E_t \rvert$	Absolute Percent Error, $(\lvert E_t \rvert / D_t)(100)$
1	200	225	−25	625	25	12.5%
2	240	220	20	400	20	8.3
3	300	285	15	225	15	5.0
4	270	290	−20	400	20	7.4
5	230	250	−20	400	20	8.7
6	260	240	20	400	20	7.7
7	210	250	−40	1,600	40	19.0
8	275	240	35	1,225	35	12.7
		Total	−15	5,275	195	81.3%

SOLUTION

Using the formulas for the measures, we get

Cumulative forecast error (bias):

CFE = −15 (the bias, or the sum of the errors for all time periods in the time series)

Average forecast error (mean bias):

$$\bar{E} = \frac{\text{CFE}}{n} = \frac{-15}{8} = -1.875$$

Mean squared error:

$$\text{MSE} = \frac{\Sigma E_t^2}{n} = \frac{5{,}275}{8} = 659.4$$

Standard deviation of the errors:

$$\sigma = \sqrt{\frac{\Sigma[E_t - (-1.875)]^2}{7}} = 27.4$$

Mean absolute deviation:

$$\text{MAD} = \frac{\Sigma |E_t|}{n} = \frac{195}{8} = 24.4$$

Mean absolute percent error:

$$\text{MAPE} = \frac{[\Sigma |E_t| / D_t]100}{n} = \frac{81.3\%}{8} = 10.2\%$$

A CFE of −15 indicates that the forecast has a slight bias to overestimate demand. The MSE, σ, and MAD statistics provide measures of forecast error variability. A MAD of 24.4 means that the average forecast error was 24.4 units in absolute value. The value of σ, 27.4, indicates that the sample distribution of forecast errors has a standard deviation of 27.4 units. A MAPE of 10.2 percent implies that, on average, the forecast error was about 10 percent of actual demand. These measures become more reliable as the number of periods of data increases.

DECISION POINT

Although reasonably satisfied with these forecast performance results, the analyst decided to test out a few more forecasting methods before reaching a final forecasting method to use for the future.

Computer Support

Computer support, such as from OM Explorer or POM for Windows, makes error calculations easy when evaluating how well forecasting models fit with past data. Errors are measured across past data, often called the *history file* in practice. They show the various error measures across the entire history file for each forecasting method evaluated. They also make forecasts into the future, based on the method selected.

Judgment Methods

Forecasts from quantitative methods are possible only when there is adequate historical data, (i.e., the *history file*). However, the history file may be nonexistent when a new product is introduced or when technology is expected to change. The history file might exist but be less useful when certain events (such as rollouts or special packages) are reflected in the past data, or when certain events are expected to occur in the future. In some cases, judgment methods are the only practical way to make a forecast. In other cases, judgment methods can also be used to modify forecasts that are generated by quantitative methods. They may recognize that one or two quantitative models have been performing particularly well in recent periods. Adjustments certainly would be called for if the forecaster has important contextual knowledge. *Contextual knowledge* is knowledge that practitioners gain through experience, such as cause-and-effect relationships, environmental cues, and organizational information that may have an effect on the variable being forecast. Adjustments also could account for unusual circumstances, such as a new sales promotion or unexpected international events. They could also have been used to remove the effect of special one-time events in the history file before quantitative methods are applied. Four of the more successful judgment methods are as follows: (1) salesforce estimates, (2) executive opinion, (3) market research, and (4) the Delphi method.

salesforce estimates

The forecasts that are compiled from estimates of future demands made periodically by members of a company's salesforce.

Salesforce estimates are forecasts compiled from estimates made periodically by members of a company's salesforce. The salesforce is the group most likely to know which services or products customers will be buying in the near future and in what quantities. Forecasts of individual salesforce members can be combined easily to get regional or national sales estimates. However, individual biases of the salespeople may taint the forecast. For example, some people are naturally optimistic, whereas others are more cautious. Adjustments in forecasts may need to be made to account for these individual biases.

executive opinion

A forecasting method in which the opinions, experience, and technical knowledge of one or more managers are summarized to arrive at a single forecast.

Executive opinion is a forecasting method in which the opinions, experience, and technical knowledge of one or more managers or customers are summarized to arrive at a single forecast. All of the factors going into judgmental forecasts would fall into the category of executive opinion. Executive opinion can also be used for **technological forecasting**. The quick pace of technological change makes keeping abreast of the latest advances difficult.

technological forecasting

An application of executive opinion to keep abreast of the latest advances in technology.

market research

A systematic approach to determine external consumer interest in a service or product by creating and testing hypotheses through data-gathering surveys.

Market research is a systematic approach to determine external consumer interest in a service or product by creating and testing hypotheses through data-gathering surveys. Conducting a market research study includes designing a questionnaire, deciding how to administer it, selecting a representative sample, and analyzing the information using judgment and statistical tools to interpret the responses. Although market research yields important information, it typically includes numerous qualifications and hedges in the findings.

Delphi method

A process of gaining consensus from a group of experts while maintaining their anonymity.

The **Delphi method** is a process of gaining consensus from a group of experts while maintaining their anonymity. This form of forecasting is useful when no historical data are available from which to develop statistical models and when managers inside the firm have no experience on which to base informed projections. A coordinator sends questions to each member of the group of outside experts, who may not even know who else is participating. The coordinator prepares a statistical summary of the responses along with a summary of arguments for particular responses. The report is sent to the same group for another round, and the participants may choose to modify their previous responses. These rounds continue until consensus is obtained.

In the remainder of this chapter, we turn to the commonly used quantitative forecasting approaches.

Causal Methods: Linear Regression

Causal methods are used when historical data are available and the relationship between the factor to be forecasted and other external or internal factors (e.g., government actions or advertising promotions) can be identified. These relationships are expressed in mathematical terms and can be complex. Causal methods are good for predicting turning points in demand and for preparing long-range forecasts. We focus on linear regression, one of the best known and most commonly used causal methods.

linear regression

A causal method in which one variable (the dependent variable) is related to one or more independent variables by a linear equation.

dependent variable

The variable that one wants to forecast.

In **linear regression**, one variable, called a dependent variable, is related to one or more independent variables by a linear equation. The **dependent variable** (such as demand for door

hinges) is the one the manager wants to forecast. The **independent variables** (such as advertising expenditures and new housing starts) are assumed to affect the dependent variable and thereby "cause" the results observed in the past. Figure 14.3 shows how a linear regression line relates to the data. In technical terms, the regression line minimizes the squared deviations from the actual data.

independent variables
Variables that are assumed to affect the dependent variable and thereby "cause" the results observed in the past.

In the simplest linear regression models, the dependent variable is a function of only one independent variable and, therefore, the theoretical relationship is a straight line:

$$Y = a + bX$$

where

$Y =$ dependent variable

$X =$ independent variable

$a = Y$-intercept of the line

$b =$ slope of the line

The objective of linear regression analysis is to find values of a and b that minimize the sum of the squared deviations of the actual data points from the graphed line. Computer programs are used for this purpose. For any set of matched observations for Y and X, the program computes the values of a and b and provides measures of forecast accuracy. Three measures commonly reported are (1) the sample correlation coefficient, (2) the sample coefficient of determination, and (3) the standard error of the estimate.

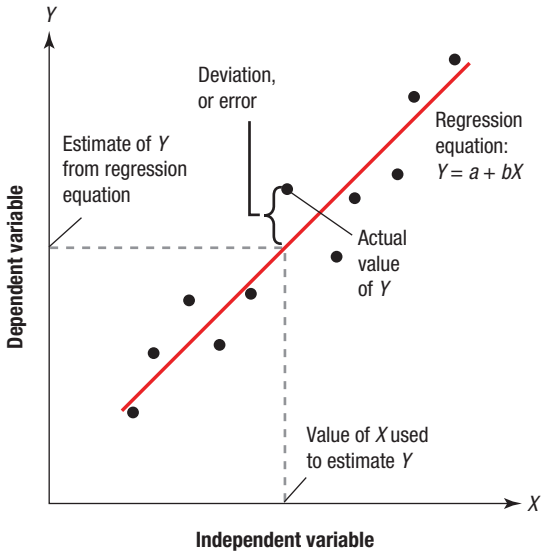


FIGURE 14.3 ▲
Linear Regression Line Relative to Actual Demand

The *sample correlation coefficient*, r, measures the direction and strength of the relationship between the independent variable and the dependent variable. The value of r can range from -1.00 to $+1.00$. A correlation coefficient of $+1.00$ implies that period-by-period changes in direction (increases or decreases) of the independent variable are always accompanied by changes in the same direction by the dependent variable. An r of -1.00 means that decreases in the independent variable are always accompanied by increases in the dependent variable, and vice versa. A zero value of r means no linear relationship exists between the variables. The closer the value of r is to $\pm$ 1.00, the better the regression line fits the points.

The *sample coefficient of determination* measures the amount of variation in the dependent variable about its mean that is explained by the regression line. The coefficient of determination is the square of the correlation coefficient, or r^2. The value of r^2 ranges from 0.00 to 1.00. Regression equations with a value of r^2 close to 1.00 mean a close fit.

The *standard error of the estimate*, s_{xy}, measures how closely the data on the dependent variable cluster around the regression line. Although it is similar to the sample standard deviation, it measures the error from the dependent variable, Y, to the regression line, rather than to the mean. Thus, it is the standard deviation of the difference between the actual demand and the estimate provided by the regression equation.

EXAMPLE 14.2 Using Linear Regression to Forecast Product Demand

The supply chain manager seeks a better way to forecast the demand for door hinges and believes that the demand is related to advertising expenditures. The following are sales and advertising data for the past 5 months:

Month	Sales (Thousands of Units)	Advertising (Thousands of $)
1	264	2.5
2	116	1.3
3	165	1.4
4	101	1.0
5	209	2.0

The company will spend $1,750 next month on advertising for the product. Use linear regression to develop an equation and a forecast for this product.

MyOMLab
Active Model 14.1 in MyOMLab provides insight on varying the intercept and slope of the model.

SOLUTION

We used POM for Windows to determine the best values of a, b, the correlation coefficient, the coefficient of determination, and the standard error of the estimate.

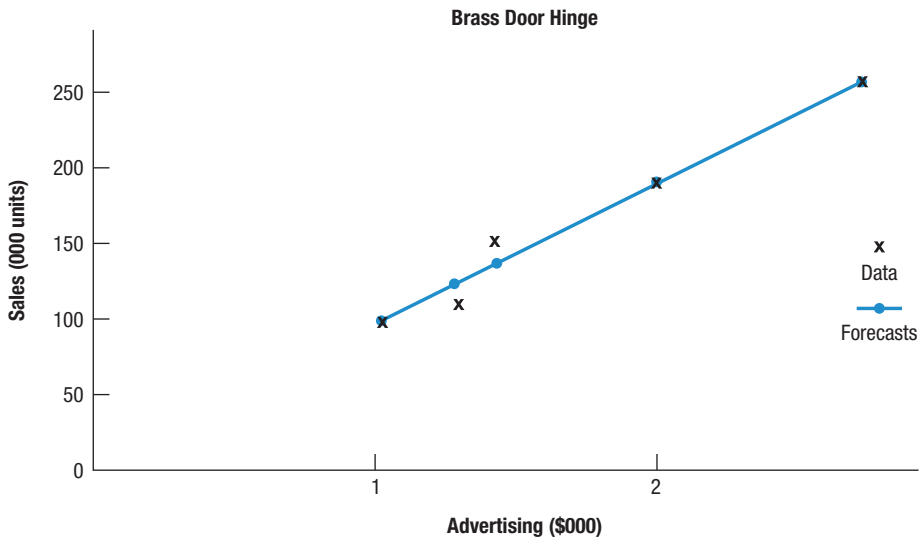
$$a = -8.135$$
$$b = 109.229X$$
$$r = 0.980$$
$$r^2 = 0.960$$
$$s_{yx} = 15.603$$

The regression equation is

$$Y = -8.135 + 109.229X$$

and the regression line is shown in Figure 14.4. The sample correlation coefficient, r, is 0.98, which is unusually close to 1.00 and suggests an unusually strong positive relationship exists between sales and advertising expenditures. The sample coefficient of determination, r^2, implies that 96 percent of the variation in sales is explained by advertising expenditures.

FIGURE 14.4 ▶
Linear Regression Line for the Sales and Advertising Data Using POM for Windows



DECISION POINT

The supply chain manager decided to use the regression model as input to planning production levels for month 6. As the advertising expenditure will be $1,750, the forecast for month 6 is $Y = -8.135 + 109.229(1.75) = 183.016$, or 183,016 units.

Often several independent variables may affect the dependent variable. For example, advertising expenditures, new corporation start-ups, and residential building contracts all may be important for estimating the demand for door hinges. In such cases, *multiple regression analysis* is helpful in determining a forecasting equation for the dependent variable as a function of several independent variables. Such models can be analyzed with POM for Windows or OM Explorer and can be quite useful for predicting turning points and solving many planning problems.

Time-Series Methods

Rather than using independent variables for the forecast as regression models do, time-series methods use historical information regarding only the dependent variable. These methods are based on the assumption that the dependent variable's past pattern will continue in the future. Time-series analysis identifies the underlying patterns of demand that combine to produce an observed historical pattern of the dependent variable and then develops a model to replicate it. In this section, we focus on time-series methods that address the horizontal, trend, and seasonal patterns of demand. Before we discuss statistical methods, let us take a look at the simplest time-series method for addressing all patterns of demand—the naïve forecast.

Naïve Forecast

A method often used in practice is the **naïve forecast**, whereby the forecast for the next period (F_{t+1}) equals the demand for the current period (D_t). So if the actual demand for Wednesday is 35 customers, the forecasted demand for Thursday is 35 customers. Despite its name, the naïve forecast can perform well.

naïve forecast
A time-series method whereby the forecast for the next period equals the demand for the current period, or Forecast = D_t

The naïve forecast method may be adapted to take into account a demand trend. The increase (or decrease) in demand observed between the last two periods is used to adjust the current demand to arrive at a forecast. Suppose that last week the demand was 120 units and the week before it was 108 units. Demand increased 12 units in 1 week, so the forecast for next week would be 120 + 12 = 132 units. The naïve forecast method also may be used to account for seasonal patterns. If the demand last July was 50,000 units, and assuming no underlying trend from one year to the next, the forecast for this July would be 50,000 units. The method works best when the horizontal, trend, or seasonal patterns are stable and random variation is small.

Estimating the Average

We begin our discussion of statistical methods of time-series forecasting with demand that has no apparent trend, seasonal, or cyclical patterns. The horizontal pattern in a time series is based on the mean of the demands, so we focus on forecasting methods that estimate the average of a time series of data. The forecast of demand for *any* period in the future is the average of the time series computed in the current period. For example, if the average of past demand calculated on Tuesday is 65 customers, the forecasts for Wednesday, Thursday, and Friday are 65 customers each day.

Consider Figure 14.5, which shows patient arrivals at a medical clinic over the past 28 weeks. Assuming that the time series has only a horizontal and random pattern, one approach is simply to calculate the average of the data. However, this approach has no adaptive quality if there is a trend, seasonal, or cyclical pattern. The statistical techniques that do have an adaptive quality in estimating the average in a time series are (1) simple moving averages, (2) weighted moving averages, and (3) exponential smoothing. Another option is the simple average, but it has no adaptive capability.

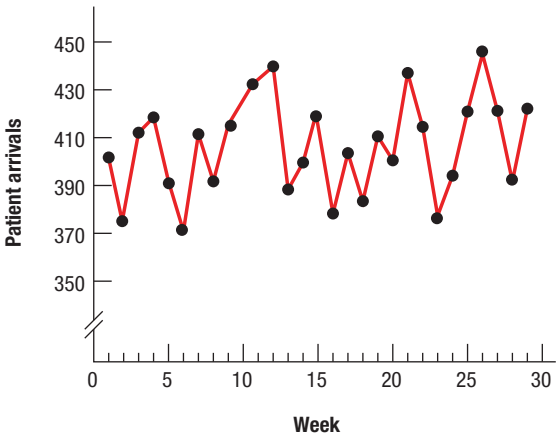


FIGURE 14.5 ▲
Weekly Patient Arrivals at a Medical Clinic

Simple Moving Averages The **simple moving average method** simply involves calculating the average demand for the n most recent time periods and using it as the forecast for future time periods. For the next period, after the demand is known, the oldest demand from the previous average is replaced with the most recent demand and the average is recalculated. In this way, the n most recent demands are used, and the average "moves" from period to period.

simple moving average method
A time-series method used to estimate the average of a demand time series by averaging the demand for the n most recent time periods.

Specifically, the forecast for period $t + 1$ can be calculated at the end of period t (after the actual demand for period t is known) as

$$F_{t+1} = \frac{\text{Sum of last } n \text{ demands}}{n} = \frac{D_t + D_{t-1} + D_{t-2} + \cdots + D_{t-n+1}}{n}$$

where

D_t = actual demand in period t
n = total number of periods in the average
F_{t+1} = forecast for period $t + 1$

EXAMPLE 14.3 Using the Moving Average Method to Estimate Average Demand

a. Compute a *three-week* moving average forecast for the arrival of medical clinic patients in week 4. The numbers of arrivals for the past 3 weeks were as follows:

Week	Patient Arrivals
1	400
2	380
3	411

b. If the actual number of patient arrivals in week 4 is 415, what is the forecast error for week 4?

c. What is the forecast for week 5?

MyOMLab
Active Model 14.2 in MyOMLab provides insight on the impact of varying n using the example in Figure 14.5.

MyOMLab
Tutor 14.1 in MyOMLab provides another example to practice making forecasts with the moving average method.

SOLUTION

a. The moving average forecast at the end of week 3 is

$$F_4 = \frac{411 + 380 + 400}{3} = 397.0$$

b. The forecast error for week 4 is

$$E_4 = D_4 - F_4 = 415 - 397 = 18$$

c. The forecast for week 5 requires the actual arrivals from weeks 2 through 4, the 3 most recent weeks of data.

$$F_5 = \frac{415 + 411 + 380}{3} = 402.0$$

DECISION POINT

Thus, the forecast at the end of week 3 would have been 397 patients for week 4, which fell short of actual demand by 18 patients. The forecast for week 5, made at the end of week 4, would be 402 patients. If a forecast is needed now for week 6 and beyond, it would also be for 402 patients.

The moving average method may involve the use of as many periods of past demand as desired. Large values of n should be used for demand series that are stable, and small values of n should be used for those that are susceptible to changes in the underlying average. If n is set to its lowest level (i.e., 1), it becomes the naïve method.

weighted moving average method
A time-series method in which each historical demand in the average can have its own weight; the sum of the weights equals 1.0.

Weighted Moving Averages In the simple moving average method, each demand has the same weight in the average—namely, $1/n$. In the **weighted moving average method**, each historical demand in the average can have its own weight. The sum of the weights equals 1.0. For example, in a *three-period* weighted moving average model, the most recent period might be assigned a weight of 0.50, the second most recent might be weighted 0.30, and the third most recent might be weighted 0.20. The average is obtained by multiplying the weight of each period by the value for that period and adding the products together:

$$F_{t+1} = 0.50D_t + 0.30D_{t-1} + 0.20D_{t-2}$$

For a numerical example of using the weighted moving average method to estimate average demand, see Solved Problem 4 and Tutor 14.2 of OM Explorer in MyOMLab.

The advantage of a weighted moving average method is that it allows you to emphasize recent demand over earlier demand. (It can even handle seasonal effects by putting higher weights on prior years in the same season.) The forecast will be more responsive to changes in the underlying average of the demand series than the simple moving average forecast.

exponential smoothing method
A weighted moving average method that calculates the average of a time series by implicitly giving recent demands more weight than earlier demands.

Exponential Smoothing The **exponential smoothing method** is a sophisticated weighted moving average method that calculates the average of a time series by implicitly giving recent demands more weight than earlier demands, all the way back to the first period in the history file. It is the most frequently used formal forecasting method because of its simplicity and the small amount of data needed to support it. Unlike the weighted moving average method, which requires n periods of past demand and n weights, exponential smoothing requires only three items of data: (1) the last period's forecast; (2) the actual demand for this period; and (3) a smoothing parameter, alpha (α), which has a value between 0 and 1.0. The equation for the exponentially smoothed forecast for period $t + 1$ is calculated

$$F_{t+1} = \alpha D_t + (1 - \alpha)F_t$$



Art Directors & TRIP/Alamy

Unilever—the purveyor of Lipton Tea, Dove, Hellmann's, and hundreds of other brands, must forecast demand around the world. It has a state-of-the-art forecasting system. Using software from Manugistics, the system blends forecasts from time series techniques with judgmental adjustments for planned promotions from its sales teams. Unilever compares point-of-sales data with its own forecasts. The forecasts are reviewed and judgmentally adjusted as needed.

The emphasis given to the most recent demand levels can be adjusted by changing the smoothing parameter. Larger α values emphasize recent levels of demand and result in forecasts more responsive to changes in the underlying average. Smaller α values treat past demand more uniformly and result in more stable forecasts. Smaller α values are analogous to increasing the value of n in the moving average method and giving greater weight to past demand. In practice, various values of α are tried and the one producing the best forecasts is chosen.

Exponential smoothing requires an initial forecast to get started. There are several ways to get this initial forecast. OM Explorer and POM for Windows use as a default setting the actual demand in the first period, which becomes the forecast for the second period. Forecasts and forecast errors then are calculated beginning with period 2. If some historical data are available, the initial forecast can be found by calculating the average of several recent periods of demand. The effect of the initial estimate of the average on successive estimates of the average diminishes over time.

EXAMPLE 14.4 Using Exponential Smoothing to Estimate Average Demand

a. Reconsider the patient arrival data in Example 14.3. It is now the end of week 3, so the actual number of arrivals is known to be 411 patients. Using $\alpha = 0.10$, calculate the exponential smoothing forecast for week 4.

b. What was the forecast error for week 4 if the actual demand turned out to be 415?

c. What is the forecast for week 5?

MyOMLab
Active Model 14.3 in MyOMLab provides insight on the impact of varying α in Figure 14.5.

MyOMLab
Tutor 14.3 in MyOMLab provides a new practice example of how to make forecasts with the exponential smoothing method.

SOLUTION

a. The exponential smoothing method requires an initial forecast. Suppose that we take the demand data for the first 2 weeks and average them, obtaining $(400 + 380)/2 = 390$ as an initial forecast. (POM for Windows and OM Explorer simply use the actual demand for the first week as a default setting for the initial forecast for period 1, and do not begin tracking forecast errors until the second period). To obtain the forecast for week 4, using exponential smoothing with $D_3 = 411$, $\alpha = 0.10$, and $F_3 = 390$, we calculate the forecast for week 4 as

$$F_4 = 0.10(411) + 0.90(390) = 392.1$$

Thus, the forecast for week 4 would be 392 patients.

b. The forecast error for week 4 is

$$E_4 = 415 - 392 = 23$$

c. The new forecast for week 5 would be

$$F_5 = 0.10(415) + 0.90(392.1) = 394.4$$

or 394 patients. Note that we used F_4, not the integer-value forecast for week 4, in the computation for F_5. In general, we round off (when it is appropriate) only the final result to maintain as much accuracy as possible in the calculations.

DECISION POINT

Using this exponential smoothing model, the analyst's forecasts would have been 392 patients for week 4 and then 394 patients for week 5 and beyond. As soon as the actual demand for week 5 is known, then the forecast for week 6 will be updated.

Because exponential smoothing is simple and requires minimal data, it is inexpensive and attractive to firms that make thousands of forecasts for each time period. However, its simplicity also is a disadvantage when the underlying average is changing, as in the case of a demand series with a trend. Like any method geared solely to the assumption of a stable average, exponential smoothing results will lag behind changes in the underlying average of demand. Higher α values may help reduce forecast errors when there is a change in the average; however, the lags will still occur if the average is changing systematically. Typically, if large α values (e.g., > 0.50) are required for an exponential smoothing application, chances are good that another model is needed because of a significant trend or seasonal influence in the demand series.

Trend Projection with Regression

Let us now consider a demand time series that has a trend. A *trend* in a time series is a systematic increase or decrease in the average of the series over time. Where a significant trend is present, forecasts from naïve, moving average, and exponential smoothing approaches are adaptive, but still lag behind actual demand and tend to be below or above the actual demand.

Trend projection with regression is a forecasting model that accounts for the trend with simple regression analysis. To develop a regression model for forecasting the trend, let the dependent variable, Y, be a period's demand and the independent variable, t, be the time period. For the first period, let $t = 1$; for the second period, let $t = 2$; and so on. The regression equation is

$$F_t = a + bt$$

One advantage of the trend projection with regression model is that it can forecast demand well into the future. The previous models project demand just one period ahead, and assume that demand beyond that will remain at that same level. Of course, all of the models (including the trend projection with regression model) can be updated each period to stay current. One *apparent* disadvantage of the trend with regression model is that it is not adaptive. The solution to this problem comes when you answer the following question. If you had the past sales of Ford automobiles since 1920, would you include each year in your regression analysis, giving equal weight to each year's sales, or include just the sales for more recent years? You most likely would decide to include just the more recent years, making your regression model more adaptive. The trend projection with regression model can thus be made more or less adaptive by the selection of historical data periods to include in the same way that moving average (changing n) or exponential smoothing (changing α) models do.

The trend projection with regression model can be solved with either the *Trend Projection with Regression* Solver or the *Time Series Forecasting* Solver in OM Explorer. Both solvers provide the regression coefficients, coefficient of determination r^2, error measures, and forecasts into the future. POM for Windows has an alternative model (we do not cover in the textbook, although a description is provided in MyOMLab) that includes the trend, called the *Trend-Adjusted Smoothing* model.

MyOMLab

The *Trend Projection with Regression* Solver focuses exclusively on trend analysis. Its graph gives a big-picture view of how well the model fits the actual demand. Its sliders allow you to control when the regression begins, how many periods are included in the regression analysis, and how many periods you want forecasted into the future. The *Time Series Forecasting* Solver, on the other hand, covers all time series models, including the trend projection with regression. It also computes a combination forecast, which we cover in a subsequent section on using multiple techniques.

EXAMPLE 14.5 Using Trend Projection with Regression to Forecast a Demand Series with a Trend

MyOMLab
Active Model 14.4 in MyOMLab provides insight on the behavior of the Trend Projetion with Regression model on the Medanalysis data.

Medanalysis, Inc., provides medical laboratory services to patients of Health Providers, a group of 10 family-practice doctors associated with a new health maintenance program. Managers are interested in forecasting the number of blood analysis requests per week. Recent publicity about the damaging effects of cholesterol on the heart has caused a national increase in requests for standard blood tests. The arrivals over the last 16 weeks are given in Table 14.1. What is the forecasted demand for the next three periods?

TABLE 14.1 ARRIVALS AT MEDANALYSIS FOR LAST 16 WEEKS

Week	Arrivals	Week	Arrivals
1	28	9	61
2	27	10	39
3	44	11	55
4	37	12	54
5	35	13	52
6	53	14	60
7	38	15	60
8	57	16	75

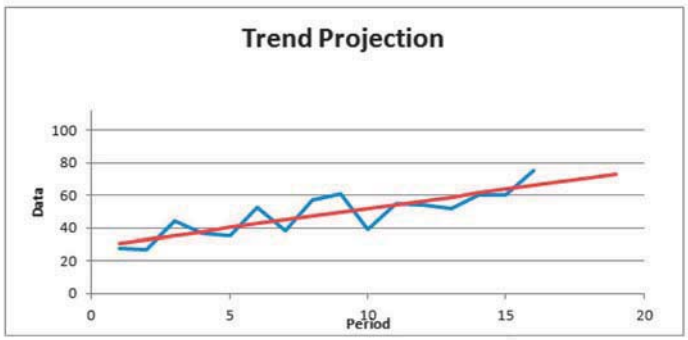
SOLUTION

Figure 14.6(a) shows the results using the *Trend Projection with Regression* Solver when all 16 weeks are included in the regression analysis, with Figure 14.6(b) showing the worksheet that goes with it.

Solver - Trend Projection with Regression

Regression begins in period	1
Error analysis begins in period	1
Number of future forecasts	3

▼ FIGURE 14.6(a)
First Model



a (Y intercept)	28.50
b (slope or trend)	2.35
r2	0.69
CFE	0.00
MAD	6.21
MSE	52.96
MAPE	13.53%
Forecast for period 17	68.375
Forecast for period 18	70.72059
Forecast for period 19	73.06618

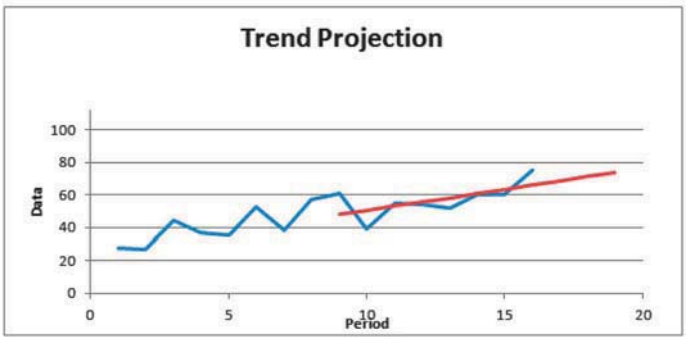
▼ FIGURE 14.6(b)
Detailed Calculations of Forecast Errors for First Model

					Averages		
				CFE	MSE	MAD	MAPE
				0.000	52.958	6.210	13.53%
Period #	Actual Demand	Forecast	Error	Running CFE	Error Squared	Absolute Error	Abs % error
1	28	31	-2.846	-2.846	8.097	2.846	10.16%
2	27	33	-6.191	-9.037	38.331	6.191	22.93%
3	44	36	8.463	-0.574	71.626	8.463	19.23%
4	37	38	-0.882	-1.456	0.779	0.882	2.38%
5	35	40	-5.228	-6.684	27.331	5.228	14.94%
6	53	43	10.426	3.743	108.711	10.426	19.67%
7	38	45	-6.919	-3.176	47.874	6.919	18.21%
8	57	47	9.735	6.559	94.776	9.735	17.08%
9	61	50	11.390	17.949	129.725	11.390	18.67%
10	39	52	-12.956	4.993	167.855	12.956	33.22%
11	55	54	0.699	5.691	0.488	0.699	1.27%
12	54	57	-2.647	3.044	7.007	2.647	4.90%
13	52	59	-6.993	-3.949	48.897	6.993	13.45%
14	60	61	-1.338	-5.287	1.791	1.338	2.23%
15	60	64	-3.684	-8.971	13.571	3.684	6.14%
16	75	66	8.971	0.000	80.471	8.971	11.96%

▼ FIGURE 14.6(c)
Second Model

Solver - Trend Projection with Regression

Regression begins in period	9
Error analysis begins in period	9
Number of future forecasts	3



a (Y intercept)	24.86
b (slope or trend)	2.57
r2	0.39
CFE	0.00
MAD	5.96
MSE	55.29
MAPE	11.10%

Forecast for period 17	68.57143
Forecast for period 18	71.14286
Forecast for period 19	73.71429

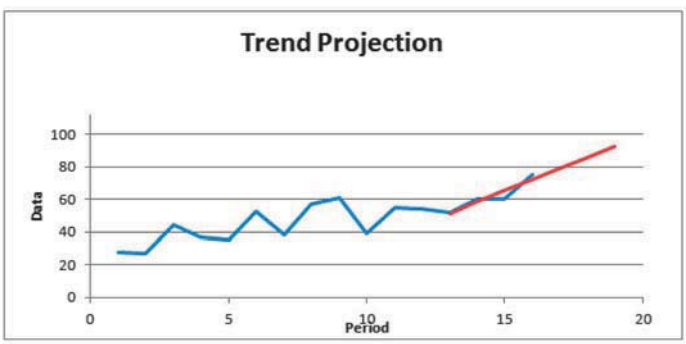
Looking at the Results sheet of Figure 14.6(a), we see that the *Y* intercept of the trend line (*a*) is 28.50 and the slope of the line (*b*) is 2.35. Thus, the trend equation is $F_t = a + bt$, where t is the time period for which you are forecasting. The forecast for period 19 is 28.5 + 2.35 (19) = 73. The error terms are CFE = 0 (which is to be expected when the regression begins at the same time that error analysis begins), MAD = 6.21, MSE = 52.96, and MAPE = 13.53 percent. The coefficient of determination r^2 is decent at 0.69. The trend line is rising gently and reaches 73 for period 19. Each period the forecast predicts an increase of 2.35 arrivals per week.

When the number of periods included in the regression analysis is reduced to 9, Figure 14.6(c) shows this second model produces mixed results. The trend line has a steeper slope. MAD and MAPE are better, but r^2 and MSE are worse. The third model in Figure 14.6(d) is the extreme, where only the last four periods are used in building the regression model. It has the best r^2, and all of the error measures are much better than the first two models. Its forecast for period 19 is 93 arrivals. However, this model is based only on the last 4 weeks of data, ignoring all previous data in the history file. For that reason, management decided to split the difference with a forecast of 83 arrivals. It is halfway between the more conservative forecast of 73 in Figure 14.6(a) and Figure 14.6(c), and the optimistic forecast of 93 in Figure 14.6(d).

Solver - Trend Projection with Regression

Regression begins in period	13
Error analysis begins in period	13
Number of future forecasts	3

FIGURE 14.6(d) ▶
Third Model



a (Y intercept)	-38.30
b (slope or trend)	6.90
r2	0.86
CFE	0.00
MAD	2.60
MSE	9.67
MAPE	4.13%

Forecast for period 17	79
Forecast for period 18	85.9
Forecast for period 19	92.8

Seasonal Patterns

Seasonal patterns are regularly repeating upward or downward movements in demand measured in periods of less than one year (hours, days, weeks, months, or quarters). In this context, the time periods are called *seasons*. For example, customer arrivals at a fast-food shop on any day may peak between 11 A.M. and 1 P.M. and again from 5 P.M. to 7 P.M.

An easy way to account for seasonal effects is to use one of the techniques already described, but to limit the data in the time series to those time periods in the same season. For example, for a day-of-the-week seasonal effect, one time series would be for Mondays, one for Tuesdays, and so on. Such an approach accounts for seasonal effects, but has the disadvantage of discarding considerable information on past demand.

Other methods are available that analyze all past data, using one model to forecast demand for all of the seasons. We describe only the **multiplicative seasonal method**, whereby an estimate of average demand is multiplied by seasonal factors to arrive at a seasonal forecast. The four-step procedure presented here involves the use of simple averages of past demand, although more sophisticated methods for calculating averages, such as a moving average or exponential smoothing approach, could be used. The following description is based on a seasonal pattern lasting one year and seasons of one month, although the procedure can be used for any seasonal pattern and season of any length.

multiplicative seasonal method
A method whereby seasonal factors are multiplied by an estimate of average demand to arrive at a seasonal forecast.

1. For each year, calculate the average demand per season by dividing annual demand by the number of seasons per year.
2. For each year, divide the actual demand for a season by the average demand per season. The result is a *seasonal index* for each season in the year, which indicates the level of demand relative to the average demand. For example, a seasonal index of 1.14 calculated for April implies that April's demand is 14 percent greater than the average demand per month.
3. Calculate the average seasonal index for each season, using the results from step 2. Add the seasonal indices for a season and divide by the number of years of data.
4. Calculate each season's forecast for next year. Begin by forecasting next year's annual demand using the naïve method, moving averages, exponential smoothing, or trend projection with regression. Then, divide annual demand by the number of seasons per year to get the average demand per season. Finally, make the seasonal forecast by multiplying the average demand per season by the appropriate seasonal index found in step 3.

EXAMPLE 14.6 Using the Multiplicative Seasonal Method to Forecast the Number of Customers

The manager of the Stanley Steemer carpet cleaning company needs a quarterly forecast of the number of customers expected next year. The carpet cleaning business is seasonal, with a peak in the third quarter and a trough in the first quarter. The manager wants to forecast customer demand for each quarter of year 5, based on an estimate of total year 5 demand of 2,600 customers.

SOLUTION

The following table calculates the seasonal factor for each week.

It shows the quarterly demand data from the past 4 years, as well as the calculations performed to get the average seasonal factor for each quarter.

	YEAR 1		YEAR 2		YEAR 3		YEAR 4		
Quarter	Demand	Seasonal Factor (1)	Demand	Seasonal Factor (2)	Demand	Seasonal Factor (3)	Demand	Seasonal Factor (4)	Average Seasonal Factor [(1+2+3+4+)/4]
1	45	45/250 = 0.18	70	70/300 = 0.23333	100	100/450 = 0.22222	100	100/550 = 0.18182	0.2043
2	335	335/250 = 1.34	370	370/300 = 1.23333	585	585/450 = 1.30	725	725/550 = 1.31818	1.2979
3	520	520/250 = 2.08	590	590/300 = 1.96667	830	830/450 = 1.84444	1160	1160/550 = 2.10909	2.0001
4	100	100/250 = 0.40	170	170/300 = 0.56667	285	285/450 = 0.63333	215	215/550 = 0.39091	0.4977
Total	1,000		1,200		1,800		2,200		
Average	1,000/4 = 250		1,200/4 = 300		1,800 = 450		2,200/4 = 550		

For example, the seasonal factor for quarter 1 in year 1 is calculated by dividing the actual demand (45) by the average demand for the whole year (1000/4 = 250). When this is done for all 4 years, we then can average the seasonal factors for quarter 1 over all 4 years. The result is a seasonal factor of 0.2043 for quarter 1.

Once seasonal factors are calculated for all four seasons (see last column in the table on the previous page), we then turn to making the forecasts for year 5. The manager suggests a forecast of 2,600 customers for the whole year, which seems reasonable given that the annual demand has been increasing by an average of 400 customers each year (from 1,000 in year 1 to 2,200 in year 4, or 1,200/3 = 400. The computed forecast demand is found by extending that trend, and projecting an annual demand in year 5 of 2,200 + 400 = 2,600 customers. (This same result is confirmed using the *Trend Projection with Regression* Solver of OM Explorer.) The quarterly forecasts are straight-forward. First, find the average demand forecast for year 5, which is 2,600/4 = 650. Then multiple this average demand by the average seasonal index, giving us

Quarter	Forecast
1	$650 \times 0.2043 = 132.795$
2	$650 \times 1.2979 = 843.635$
3	$650 \times 2.0001 = 1{,}300.065$
4	$650 \times 0.4977 = 323.505$

Figure 14.7 shows the computer solution using the *Seasonal Forecasting* Solver in OM Explorer. Figure 14.7(b) confirms all of the calculations made above. Notice in Figure 14.7(a) that a computer demand forecast is provided as a default for year 5. However, there is an option for user-supplied demand forecast that overrides the computer-supplied forecast if the manager wishes to make a judgmental forecast based on additional information.

FIGURE 14.7 ▶
Demand Forecasts Using the *Seasonal Forecasting* Solver of *OM Explorer*

Period: Quarters

Starting Year: 1 Years: 4

Computed Forecast Demand for Year 5: 2600

User-supplied Forecast Demand for Year 5: 2600

	Year			
Quarter	1	2	3	4
1	45	70	100	100
2	335	370	585	725
3	520	590	830	1160
4	100	170	285	215

(a) Inputs sheet

Quarter	Seasonal Index	Forecast
1	0.2043	132.795
2	1.2979	843.635
3	2.0001	1300.065
4	0.4977	323.505

(b) Results

DECISION POINT

Using this seasonal method, the analyst makes a demand forecast as low as 133 customers in the first quarter and as high as 1,300 customers in the third quarter. The season of the year clearly makes a difference.

An alternative to the multiplicative seasonal method is the **additive seasonal method**, whereby seasonal forecasts are generated by adding or subtracting a seasonal constant (say, 50 units) to the estimate of average demand per season. This approach is based on the assumption that the seasonal pattern is constant, regardless of average demand. The amplitude of the seasonal adjustment remains the same regardless of the level of demand.

additive seasonal method
A method in which seasonal forecasts are generated by adding a constant to the estimate of average demand per season.

Choosing a Quantitative Forecasting Method

Criteria for Selecting Time-Series Methods

Forecast error measures provide important information for choosing the best forecasting method for a service or product. They also guide managers in selecting the best values for the parameters needed for the method: n for the moving average method, the weights for the weighted moving average method, α for the exponential smoothing method, and when regression data begins for the trend projection with regression method. The criteria to use in making forecast method and parameter choices include (1) minimizing bias (CFE); (2) minimizing MAPE, MAD, or MSE; (3) maximizing r^2; (4) meeting managerial expectations of changes in the components of demand; and (5) minimizing the forecast errors in recent periods. The first three criteria relate to statistical measures based on historical performance, the fourth reflects expectations of the future that may not be rooted in the past, and the fifth is a way to use whatever method seems to be working best at the time a forecast must be made.

Using Statistical Criteria Statistical performance measures can be used in the selection of which forecasting method to use. The following guidelines will help when searching for the best time-series models:

1. For projections of more stable demand patterns, use lower α values or larger n values to emphasize historical experience.
2. For projections of more dynamic demand patterns using the models covered in this chapter, try higher α values or smaller n values. When historical demand patterns are changing, recent history should be emphasized.

Often, the forecaster must make trade-offs between bias (CFE) and the measures of forecast error dispersion (MAPE, MAD, and MSE). Managers also must recognize that the best technique in explaining the past data is not necessarily the best technique to predict the future, and that "overfitting" past data can be deceptive. Such was the case in Example 14.5. All of the forecast error measures suggested that the regression model in Figure 14.5(d) was best, but management was hesitant because it used so little of the time series. A forecasting method may have small errors relative to the history file, but may generate high errors for future time periods. For this reason, some analysts prefer to use a **holdout sample** as a final test (see the two Experiential Learning Exercises at the end of this chapter). To do so, they set aside some of the more recent periods from the time series and use only the earlier time periods to develop and test different models. Once the final models have been selected in the first phase, they are tested again with the holdout sample. Performance measures, such as MAD and CFE, would still be used but they would be applied to the holdout sample. Whether this idea is used or not, managers should monitor future forecast errors, and modify their forecasting approaches as needed. Maintaining data on forecast performance is the ultimate test of forecasting power—rather than how well a model fits past data or holdout samples.

holdout sample
Actual demands from the more recent time periods in the time series that are set aside to test different models developed from the earlier time periods.

Tracking Signals

A **tracking signal** is a measure that indicates whether a method of forecasting is accurately predicting actual changes in demand. The tracking signal measures the number of MADs represented by the cumulative sum of forecast errors, the CFE. The CFE tends to be close to 0 when a correct forecasting system is being used. At any time, however, random errors can cause the CFE to be a nonzero number. The tracking signal formula is

$$\text{Tracking signal} = \frac{\text{CFE}}{\text{MAD}} \quad \text{or} \quad \frac{\text{CFE}}{\text{MAD}_t}$$

tracking signal
A measure that indicates whether a method of forecasting is accurately predicting actual changes in demand.

Each period, the CFE and MAD are updated to reflect current error, and the tracking signal is compared to some predetermined limits. The MAD can be calculated in one of two ways: (1) as the simple average of all absolute errors (as demonstrated in Example 14.1) or (2) as a weighted average determined by the exponential smoothing method:

$$\text{MAD}_t = \alpha|E_t| + (1 - \alpha)\text{MAD}_{t-1}$$

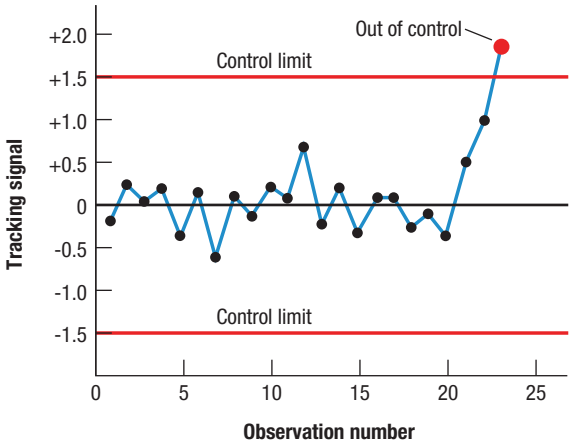


FIGURE 14.8 ▲
Tracking Signal

If forecast errors are normally distributed with a mean of 0, the relationship between σ and MAD is simple:

$$\sigma = (\sqrt{\pi/2})(\text{MAD}) \cong 1.25(\text{MAD})$$

$$\text{MAD} = 0.7978\sigma \cong 0.8\sigma$$

where

$$\pi = 3.1416$$

This relationship allows use of the normal probability tables to specify limits for the tracking signal. If the tracking signal falls outside those limits, the forecasting model no longer is tracking demand adequately. A tracking system is useful when forecasting systems are computerized because it alerts analysts when forecasts are getting far from desirable limits. Figure 14.8 shows tracking signal results for 23 periods plotted on a *control chart*. The control chart is useful for determining whether any action needs to be taken to improve the forecasting model. In the example, the first 20 points cluster around 0, as we would expect if the forecasts are not biased. The CFE will tend toward 0. When the underlying characteristics of demand change but the forecasting model does not, the tracking signal eventually goes out of control. The steady increase after the 20th point in Figure 14.8 indicates that the process is going out of control. The 21st and 22nd points are acceptable, but the 23rd point is not.

Using Multiple Techniques

We described several individual forecasting methods and showed how to assess their forecast performance. However, we need not rely on a single forecasting method. Several different forecasts can be used to arrive at a final forecast. Initial statistical forecasts using several time-series methods and regression are distributed to knowledgeable individuals, such as marketing directors and sales teams, (and sometimes even suppliers and customers) for their adjustments. They can account for current market and customer conditions that are not necessarily reflected in past data. Multiple forecasts may come from different sales teams, and some teams may have a better record on forecast errors than others.

combination forecasts

Forecasts that are produced by averaging independent forecasts based on different methods, different sources, or different data.

Research during the last two decades suggests that combining forecasts from multiple sources often produces more accurate forecasts. **Combination forecasts** are forecasts that are produced by averaging independent forecasts based on different methods, different sources, or different data. It is intriguing that combination forecasts often perform better over time than even the *best* single forecasting procedure. For example, suppose that the forecast for the next period is 100 units from technique 1 and 120 units from technique 2 and that technique 1 has provided more accurate forecasts to date. The combination forecast for next period, giving equal weight to each technique, is 110 units (or $0.5 \times 100 + 0.5 \times 120$). When this averaging technique is used consistently into the future, its combination forecasts often will be much more accurate than those of any single best forecasting technique (in this example, technique 1). Combining is most effective when the individual forecasts bring different kinds of information into the forecasting process. Forecasters have achieved excellent results by weighting forecasts equally, and this is a good starting point. However, unequal weights may provide better results under some conditions.

OM Explorer and POM for Windows allow you to evaluate several forecasting models, and then you can create combination forecasts from them. In fact, the *Time-Series Forecasting* Solver of OM Explorer automatically computes a combination forecast as a weighted average, using the weights that you supply for the various models that it evaluates. The models include the naïve, moving average, exponential smoothing, and regression projector methods. Alternately, you can create a simple Excel spreadsheet that combines forecasts generated by POM for Windows to create combination forecasts. The *Time Series Forecasting* Solver also allows you evaluate your forecasting process with a holdout sample. The forecaster makes a forecast just one period ahead, and learns of given actual demand. Next the solver computes forecasts and forecast errors for the period. The process continues to the next period in the holdout sample with the forecaster committing to a forecast for the next period. To be informed, the forecaster should also be aware of how well the other forecasting methods have been performing, particularly in the recent past.

MANAGERIAL PRACTICE 14.1 Combination Forecasts and the Forecasting Process

Fiskars Brands, Inc., totally overhauled its forecasting process. It serves 2,000 customers ranging from large discounters to local craft stores providing about 2,300 finished SKUs. Its parent company, Fiskars Corporation, is the second oldest incorporated entity in the world and produces a variety of high-quality products such as garden shears, pruners, hand tools, scissors for preschoolers, ratchet tools, screwdrivers, and the like. Business is highly seasonal and prices quite variable. About 10 percent to 15 percent of the annual revenue comes from one-time promotions, and 25 percent to 35 percent of its products are new every year.

It introduced a statistical-based analysis along with a Web-based business intelligence tool for reporting. It put much more emphasis on combination forecasts. Instead of asking members of the sales staff to provide their own forecasts, forecasts were sent to them, and they were asked for their validation and refinement. Their inputs are most useful relative to additions, deletions, and promotions. Converting multiple forecasts into one number (forecasts from time-series techniques, sales input, and customer input) creates more accurate forecasts by SKU. Fiskars's software has the ability to weigh each input. It gives more weight to a statistical forecast for in-line items, and inputs from the sales staff get much more weight for promoted products and new items.

It also segments SKUs by value and forecastability so as to focus forecasting efforts on SKUs that have the biggest impact on the business. High-value items ("A" items identified with ABC analysis in Chapter 9, "Managing Inventories") that also have high forecastability (stable demand with low forecast errors to date) tend to do well with the time-series techniques, and **judgmental adjustments** are made with caution. High-value items with low forecastability get top priority in the forecasting effort, such as with CPFR. Much less attention is given to improving forecasts for "C" items for which there is some history and fairly steady demand.

Finally, Fiskars instituted a Web-based program that gives the entire company visibility to forecast information in whatever form it needs. For example, Finance wants monthly, quarterly, and yearly projections in dollars, whereas Operations wants projections in units as well as accuracy measures. Everybody can track updated forecast information by customer, brand, and SKU.



HANDOUT/MCT/Newscom

Fiskars Brands, Inc., totally overhauled its forecasting process. Its products include garden shears, pruners, hand tools, scissors, ratchet tools, and the like. It introduced time-series tools, with much emphasis placed on combination forecasts. Instead of asking members of the sales staff to provide their own forecasts, forecasts were sent to them, and they were asked for validation and refinement. Their judgmental inputs provide valuable information relative to additions, deletions, and promotions. Combining multiple forecasts (forecasts from several time-series techniques and judgment inputs) into one number creates more accurate forecasts by SKU.

Source: David Montgomery, "Flashpoints for Changing Your Forecasting Process," *The Journal of Business Forecasting* (Winter 2006–2007), pp. 35–37; **http://www.fiskars.com**, May 21, 2011.

Another way to take advantage of multiple techniques is **focus forecasting**, which selects the best forecast (based on past error measures) from a group of forecasts generated by individual techniques. Every period, all techniques are used to make forecasts for each item. The forecasts are made with a computer because there can be 100,000 SKUs at a company, each needing to be forecast. Using the history file as the starting point for each method, the computer generates forecasts for the current period. The forecasts are compared to actual demand, and the method that produces the forecast with the least error is used to make the forecast for the next period. The method used for each item may change from period to period.

judgmental adjustment

An adjustment made to forecasts from one or more quantitative models that accounts for recognizing which models are performing particularly well in recent past, or take into account contextual information.

focus forecasting

A method of forecasting that selects the best forecast from a group of forecasts generated by individual techniques.

Putting It All Together: Forecasting as a Process

Often companies must prepare forecasts for hundreds or even thousands of services or products repeatedly. For example, a large network of health care facilities must calculate demand forecasts for each of its services for every department. This undertaking involves voluminous data that must be manipulated frequently. However, software such as Motorola Mobility's system can ease the burden of making these forecasts and coordinating the forecasts between customers



FRED GREAVES Feature Photo Service/Newscom

West Marine acquired its East Coast competitor, E&B Marine, in 1997. The consequences were quickly apparent. Peak-season out-of-stock levels rose more than 12 percent compared to the prior year. After six years of steady growth, net income dropped from $15 million in 1997 to not much more than $1 million the next year. Fast-forward six years. They had no supply problems in any of their warehouses or stores. What changed? Two words: supply chain. Managers recognized that they needed to make a significant shift in managing its supply chain. A crucial element was greater collaboration with their suppliers. It is not enough to coordinate the supply chain within the boundaries of a single organization.

and suppliers. Many forecasting software packages are available, including Manugistics, Forecast Pro, and SAS. The forecasting routines in OM Explorer and POM for Windows give some hint of their capabilities. Forecasting is not just a set of techniques, but instead a process that must be designed and managed. While there is no one process that works for everyone, here we describe two comprehensive processes that can be quite effective in managing operations and the supply chain.

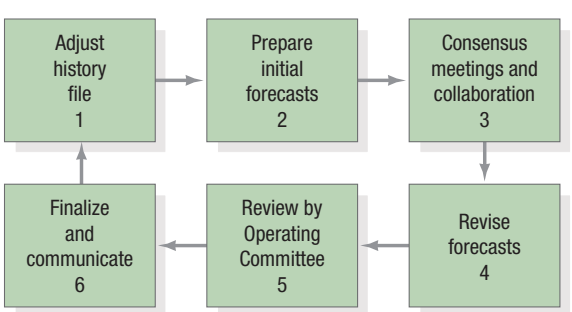
A Typical Forecasting Process

Many *inputs* to the forecasting process are informational, beginning with the *history file* on past demand. The history file is kept up-to-date with the actual demands. Clarifying notes and adjustments are made to the database to explain unusual demand behavior, such as the impact of special promotions and closeouts. Often the database is separated into two parts: *base* data and *nonbase* data. The second category reflects irregular demands. Final forecasts just made at the end of the prior cycle are entered in the history file, so as to track forecast errors. Other information sources are from salesforce estimates, outstanding bids on new orders, booked orders, market research studies, competitor behavior, economic outlook, new product introductions, pricing, and promotions. If CPFR is used, as is done by Motorola Mobility in our opening vignette, then considerable information sharing will take place with customers and suppliers. For new products, a history database is fabricated based on the firm's experience with prior products and the judgment of personnel.

Outputs of the process are forecasts for multiple time periods into the future. Typically, they are on a monthly basis and are projected out from six months to two years. Most software packages have the ability to "roll up" or "aggregate" forecasts for individual stock-keeping units (SKUs) into forecasts for whole product families. Forecasts can also be "blown down" or "disaggregated" into smaller pieces. In a make-to-stock environment, forecasts tend to be more detailed and can get down to specific individual products. In a make-to-order environment, the forecasts tend to be for groups of products. Similarly, if the lead times to buy raw materials and manufacture a product or provide a service are long, the forecasts go farther out into the future.

The forecast process itself, typically done on a monthly basis, consists of structured steps. These steps often are facilitated by someone who might be called a demand manager, forecast analyst, or demand/supply planner. However, many other people are typically involved before the plan for the month is authorized.

Step 1. The cycle begins mid-month just after the forecasts have been finalized and communicated to the stakeholders. Now is the time to update the history file and review forecast accuracy. At the end of the month, enter actual demand and review forecast accuracy.



Step 2. Prepare initial forecasts using some forecasting software package and judgment. Adjust the parameters of the software to find models that fit the past demand well and yet reflect the demand manager's judgment on irregular events and information about future sales pulled from various sources and business units.

Step 3. Hold consensus meetings with the stakeholders, such as marketing, sales, supply chain planners, and finance. Make it easy for business unit and field sales personnel to make inputs. Use the Internet to get collaborative information from key customers and suppliers. The goal is to arrive at consensus forecasts from all of the important players.

Step 4. Revise the forecasts using judgment, considering the inputs from the consensus meetings and collaborative sources.

Step 5. Present the forecasts to the operating committee for review and to reach a final set of forecasts. It is important to have a set of forecasts that everybody agrees upon and will work to support.

Step 6. Finalize the forecasts based on the decisions of the operating committee and communicate them to the important stakeholders. Supply chain planners are usually the biggest users.

As with all work activity, forecasting is a process and should be continually reviewed for improvements. A better process will foster better relationships between departments such as marketing, sales, and operations. It will also produce better forecasts. This principle is the first one in Table 14.2 to guide process improvements.

Adding Collaboration to the System

This process is similar to the first one, except that it adds considerable collaboration with the company's customers and suppliers, particularly in step 3. **Collaborative planning, forecasting, and replenishment (CPFR)** is a specific nine-step process for supply chain integration that allows a supplier and its customers to collaborate on making the forecast by using the Internet. Many other firms, including Motorola as described in the opening vignette, are turning to CPFR to coordinate up and down the supply chain.

Forecasting as a Nested Process

Forecasting is not a stand-alone activity, but instead part of a larger process that encompasses the remaining chapters. After all, demand is only half of the equation—the other half is supply. Future plans must be developed to supply the resources needed to meet the forecasted demand. Resources include the workforce, materials, inventories, dollars, and equipment capacity. Making sure that demand and supply plans are in balance begins in the next chapter, Chapter 15, "Planning and Scheduling Operations" and continues with Chapter 16, "Planning Sufficient Resources."



Mark Peterson/CORBIS

Walmart has long been known for its careful analysis of cash register receipts and for working with suppliers to reduce inventories. In the past, like many other retailers, Walmart did not share its forecasts with its suppliers. The result was forecast errors as much as 60 percent of actual demand. Retailers ordered more than they needed, and suppliers produced more than they could sell. To combat the ill effects of forecast errors on inventories, Benchmarking Partners, Inc., was funded in the mid-1990s by Walmart, IBM, SAP, and Manugistics to develop a software package. Walmart initiated this new approach with Listerine, a primary product of Warner-Lambert (now produced and distributed by Johnson & Johnson). The system worked in the following way during this pilot period. Walmart and Warner-Lambert independently calculated the demand they expected for Listerine six months into the future, taking into consideration factors such as past sales trends and promotion plans. They then exchanged their forecasts over the Internet. If the forecasts differed by more than a predetermined percentage, the retailer and the manufacturer used the Internet to exchange written comments and supporting data. The parties went through as many cycles as needed to converge on an acceptable forecast. They passed the pilot period in flying colors. Benefits to Walmart included a reduction in stockouts from 15 percent to 2 percent as well as significant increases in sales and reductions in inventory costs. Likewise, Warner-Lambert benefited by having a smoother production plan and lower average costs. This system was later generalized and dubbed CPFR, which stands for collaborative planning, forecasting, and replenishment.

TABLE 14.2 | SOME PRINCIPLES FOR THE FORECASTING PROCESS

- Better processes yield better forecasts.
- Demand forecasting is being done in virtually every company, either formally or informally. The challenge is to do it well—better than the competition.
- Better forecasts result in better customer service and lower costs, as well as better relationships with suppliers and customers.
- The forecast can and must make sense based on the big picture, economic outlook, market share, and so on.
- The best way to improve forecast accuracy is to focus on reducing forecast error.
- Bias is the worst kind of forecast error; strive for zero bias.
- Whenever possible, forecast at more aggregate levels. Forecast in detail only where necessary.
- Far more can be gained by people collaborating and communicating well than by using the most advanced forecasting technique or model.

Source: Based on Thomas F. Wallace and Robert A. Stahl, *Sales Forecasting: A New Approach* (Cincinnati, OH: T. E. Wallace & Company, 2002), p. 112. Copyright © 2002 T.E. Wallace & Company. Used with permission.

collaborative planning, forecasting, and replenishment (CPFR)

A nine-step process for supply chain integration that allows a supplier and its customers to collaborate on making the forecast by using the Internet.

LEARNING GOALS IN REVIEW

1. **Identify the various forecasting methods available to forecasting systems.** The section "Choosing the Type of Forecasting Technique," on pp. 486–487, gives a quick introduction to the four groups of forecasting techniques. Each is described fully in subsequent sections.
2. **Identify the various measures of forecast errors.** Review the "Forecast Error," pp. 487–489, and "Choosing a Quantitative Forecasting Method," pp. 501–502, to understand CFE, MSE, σ, MAD, MAPE, and the tracking signals.
3. **Use regression to make forecasts with one or more independent variables.** The "Causal Methods: Linear Regression" section and Example 14.2, pp. 490–492, describe how linear regression, when historical data is available, can express demand as a linear function of one or more independent variables. Example 14.2 on pp. 491–492 and Solved Problem 3 illustrate the computer output, including the various statistics on how well the regression equation fits the data.
4. **Make forecasts using the most common approaches for time-series analysis.** The "Time-Series Methods" on pp. 492–501 explain the naïve method, simple moving average, weighted moving average, and exponential smoothing techniques that are used. Examples 14.3 and 14.4 demonstrate some of the methods, as do Solved Problems 4 and 1.
5. **Make forecasts using Trend Projection with Regression.** We cover this technique on pp. 496–498, with four figures illustrating the computer output and how varying number of periods included in the regression analysis can impact the results.
6. **Describe a typical forecasting process used by businesses.** See the section "A Typical Forecasting Process,"pp. 504–505, and the six steps involved. There is much more complexity when you realize the number of SKUs involved and the need to update the history file.
7. **Explain collaborative planning, forecasting, and replenishment (CPFR).** The "Adding Collaboration to the System" section, p. 505, is a big step in increasing the coordination up and down the supply chain. In the chapter opener, we see what Motorola Mobility has done with its customers to improve its demand forecasts.

MyOMLab helps you develop analytical skills and assesses your progress with multiple problems on moving average, mean absolute deviation, mean absolute percent error, mean squared error (MSE), exponential smoothing, MAD, MAPE, multiplicative seasonal method, least squares regression model, and trend projection with regression.

MyOMLab Resources	Titles	Link to the Book
Video	*Forecasting and Supply Chain Management at Deckers Outdoor Corporation*	Using Multiple Techniques; Putting It All Together: Forecasting as a Process
Active Model Exercises	14.1 Linear Regression	Casual Methods: Linear Regression; Example 14.2 (pp. 491–492); Solved Problem 3
	14.2 Simple Moving Averages	Estimating the Average; Example 14.3 (pp. 493–494); Solved Problem 4
	14.3 Exponential Smoothing	Exponential Smoothing; Example 14.4 (p. 495); Solved Problem 1
OM Explorer Solvers	Regression Analysis	Casual Methods: Linear Regression; Example 14.2 (pp. 491–492); Solved Problem 3
	Seasonal Forecasting	Seasonal Patterns; Example 14.6 (pp. 499–500); Solved Problem 2
	Time Series Forecasting	Time Series Methods Examples 14.3–14.5
	Trend Projection with Regression	Trend Projection with Regression; Example 14.5 (pp. 496–498)
OM Explorer Tutors	14.1 Moving Average Method	Estimating the Average; Example 14.3 (pp. 493–494); Solved Problem 4
	14.2 Weighted Moving Average Method	Weighted Moving Average and Solved Problem 4
	14.3 Exponential Smoothing	Exponential Smoothing; Example 14.4 (p. 495); Solved Problem 1
Virtual Tours	Ferrara Pan	Key Decisions on Making Forecasts
	Cape Cod Chips	Demand Patterns; Judgment Methods
POM for Windows	Time Series Analysis	Time Series Methods; Examples 14.3–14.5; Seasonal Patterns; Example 14.6 (pp. 499–500); Solved Problem 2
	Regression Projector	Casual Methods: Linear Regression; Example 14.2 (pp. 491–492); Solved Problem 3
	Least Squares – Simple and Multiple Regression	Casual Methods: Linear Regression; Example 14.2 (pp. 491–492); Solved Problem 3
	Error Analysis	Judgment Methods; Forecast Error, Example 14.1 (p. 489), Choosing a Quantitative Forecasting Method; Solved Problem 1

MyOMLab Resources	Titles	Link to the Book
Student Data File	Experiential Exercise Two	Time-Series Methods; Choosing a Time-Series Method; Using Multiple Techniques
Tutorial	Trend-Adjusted Exponential Smoothing	Trend Projection with Regression (pp. 496–498)
Internet Exercise	National Climate Data Center	Casual Methods: Linear Regression; Example 14.2 (pp. 491–492); Trend Projection with Regression; Example 14.5 (pp. 496–498)
Key Equations		
Image Library		

Key Equations

1. Forecast error:

$$E_t = D_t - F_t$$

$$\text{CFE} = \Sigma E_t$$

$$\bar{E} = \frac{\text{CFE}}{n}$$

$$\text{MSE} = \frac{\Sigma E_t^2}{n}$$

$$\sigma = \sqrt{\frac{\Sigma (E_t - \bar{E})^2}{n-1}}$$

$$\text{MAD} = \frac{\Sigma |E_t|}{n}$$

$$\text{MAPE} = \frac{(\Sigma |E_t|/D_t)(100\%)}{n}$$

2. Linear regression:

$$Y = a + bX$$

3. Naïve forecasting:

$$\text{Forecast} = D_t$$

4. Simple moving average:

$$F_{t+1} = \frac{D_t + D_{t-1} + D_{t-2} + \cdots + D_{t-n+1}}{n}$$

5. Weighted moving average:

$$F_{t+1} = \text{Weight}_1(D_t) + \text{Weight}_2(D_{t-1}) + \text{Weight}_3(D_{t-2}) + \cdots + \text{Weight}_n(D_{t-n+1})$$

6. Exponential smoothing:

$$F_{t+1} = \alpha D_t + (1 - \alpha) F_t$$

7. Trend Projection using Regression

$$F_t = a + bt$$

8. Tracking signal:

$$\frac{\text{CFE}}{\text{MAD}} \text{ or } \frac{\text{CFE}}{\text{MAD}_t}$$

9. Exponentially smoothed error:

$$\text{MAD}_t = \alpha |E_t| + (1 - \alpha)\text{MAD}_{t-1}$$

Key Terms

additive seasonal method 501
aggregation 486
causal methods 487
collaborative planning, forecasting, and replenishment (CPFR) 505
combination forecasts 502
cumulative sum of forecast errors (CFE) 487
dependent variable 490
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Solved Problem 1

The monthly demand for units manufactured by the Acme Rocket Company has been as follows:

Month	Units	Month	Units
May	100	September	105
June	80	October	110
July	110	November	125
August	115	December	120

a. Use the exponential smoothing method to forecast the number of units for June to January. The initial forecast for May was 105 units; $\alpha = 0.2$.

b. Calculate the absolute percentage error for each month from June through December and the MAD and MAPE of forecast error as of the end of December.

c. Calculate the tracking signal as of the end of December. What can you say about the performance of your forecasting method?

SOLUTION

a.

Current Month, t	Calculating Forecast for Next Month $F_{t+1} = \alpha D_t + (1 - \alpha)F_t$	Forecast for Month $t + 1$
May	0.2(100) + 0.8(105) = 104.0 or 104	June
June	0.2(80) + 0.8(104.0) = 99.2 or 99	July
July	0.2(110) + 0.8(99.2) = 101.4 or 101	August
August	0.2(115) + 0.8(101.4) = 104.1 or 104	September
September	0.2(105) + 0.8(104.1) = 104.3 or 104	October
October	0.2(110) + 0.8(104.3) = 105.4 or 105	November
November	0.2(125) + 0.8(105.4) = 109.3 or 109	December
December	0.2(120) + 0.8(109.3) = 111.4 or 111	January

b.

Month, t	Actual Demand, D_t	Forecast, F_t	Error, $E_t = D_t - F_t$	Absolute Error, $\lvert E_t \rvert$	Absolute Percentage Error, $(\lvert E_t \rvert / D_t)(100\%)$
June	80	104	−24	24	30.0%
July	110	99	11	11	10.0
August	115	101	14	14	12.0
September	105	104	1	1	1.0
October	110	104	6	6	5.5
November	125	105	20	0	16.0
December	120	109	11	11	9.2
Total	765		39	87	83.7%

$$\text{MAD} = \frac{\Sigma|E_t|}{n} = \frac{87}{7} = 12.4 \text{ and MAPE} = \frac{(\Sigma|E_t|/D_t)(100)}{n} = \frac{83.7\%}{7} = 11.96\%$$

c. As of the end of December, the cumulative sum of forecast errors (CFE) is 39. Using the mean absolute deviation calculated in part (b), we calculate the tracking signal:

$$\text{Tracking signal} = \frac{\text{CFE}}{\text{MAD}} = \frac{39}{12.4} = 3.14$$

The probability that a tracking signal value of 3.14 could be generated completely by chance is small. Consequently, we should revise our approach. The long string of forecasts lower than actual demand suggests use of a trend method.

Solved Problem 2

The Northville Post Office experiences a seasonal pattern of daily mail volume every week. The following data for two representative weeks are expressed in thousands of pieces of mail:

Day	Week 1	Week 2
Sunday	5	8
Monday	20	15
Tuesday	30	32
Wednesday	35	30
Thursday	49	45
Friday	70	70
Saturday	15	10
Total	224	210

a. Calculate a seasonal factor for each day of the week.

b. If the postmaster estimates 230,000 pieces of mail to be sorted next week, forecast the volume for each day of the week.

SOLUTION

a. Calculate the average daily mail volume for each week. Then, for each day of the week, divide the mail volume by the week's average to get the seasonal factor. Finally, for each day, add the two seasonal factors and divide by 2 to obtain the average seasonal factor to use in the forecast (see part [b]).

	WEEK 1		WEEK 2		
Day	Mail Volume	Seasonal Factor (1)	Mail Volume	Seasonal Factor (2)	Average Seasonal Factor [(1) + (2)]/2
Sunday	5	5/32 = 0.15625	8	8/30 = 0.26667	0.21146
Monday	20	20/32 = 0.62500	15	15/30 = 0.50000	0.56250
Tuesday	30	30/32 = 0.93750	32	32/30 = 1.06667	1.00209
Wednesday	35	35/32 = 1.09375	30	30/30 = 1.00000	1.04688
Thursday	49	49/32 = 1.53125	45	45/30 = 1.50000	1.51563
Friday	70	70/32 = 2.18750	70	70/30 = 2.33333	2.26042
Saturday	15	15/32 = 0.46875	10	10/30 = 0.33333	0.40104
Total	224		210		
Average	224/7 = 32		210/7 = 30		

b. The average daily mail volume is expected to be 230,000/7 = 32,857 pieces of mail. Using the average seasonal factors calculated in part (a), we obtain the following forecasts:

Day	Calculation		Forecast
Sunday	0.21146(32,857) =		6,948
Monday	0.56250(32,857) =		18,482
Tuesday	1.00209(32,857) =		32,926
Wednesday	1.04688(32,857) =		34,397
Thursday	1.51563(32,857) =		49,799
Friday	2.26042(32,857) =		74,271
Saturday	0.40104(32,857) =		13,177
		Total	230,000

Solved Problem 3

Chicken Palace periodically offers carryout five-piece chicken dinners at special prices. Let Y be the number of dinners sold and X be the price. Based on the historical observations and calculations in the following table, determine the regression equation, correlation coefficient, and coefficient of determination. How many dinners can Chicken Palace expect to sell at $3.00 each?

Observation	Price (X)	Dinners Sold (Y)
1	$ 2.70	760
2	$ 3.50	510
3	$ 2.00	980
4	$ 4.20	250
5	$ 3.10	320
6	$ 4.05	480
Total	$19.55	3,300
Average	$ 3.258	550

SOLUTION

We use the computer (*Regression Analysis* Solver of OM Explorer or *Regression Projector* module of POM for Windows) to calculate the best values of a, b, the correlation coefficient, and the coefficient of determination.

$$a = 1{,}454.60$$
$$b = -277.63$$
$$r = -0.84$$
$$r^2 = 0.71$$

The regression line is

$$Y = a + bX = 1{,}454.60 - 277.63X$$

The correlation coefficient ($r = -0.84$) shows a negative correlation between the variables. The coefficient of determination ($r^2 = 0.71$) is not too large, which suggests that other variables (in addition to price) might appreciably affect sales.

If the regression equation is satisfactory to the manager, estimated sales at a price of $3.00 per dinner may be calculated as follows:

$$Y = a + bX = 1{,}454.60 - 277.63(3.00)$$
$$= 621.71 \text{ or } 622 \text{ dinners}$$

Solved Problem 4

The Polish General's Pizza Parlor is a small restaurant catering to patrons with a taste for European pizza. One of its specialties is Polish Prize pizza. The manager must forecast weekly demand for these special pizzas so that he can order pizza shells weekly. Recently, demand has been as follows:

Week	Pizzas	Week	Pizzas
June 2	50	June 23	56
June 9	65	June 30	55
June 16	52	July 7	60

a. Forecast the demand for pizza for June 23 to July 14 by using the simple moving average method with $n = 3$. Then, repeat the forecast by using the weighted moving average method with $n = 3$ and weights of 0.50, 0.30, and 0.20, with 0.50 applying to the most recent demand.

b. Calculate the MAD for each method.

SOLUTION

a. The simple moving average method and the weighted moving average method give the following results:

Current Week	Simple Moving Average Forecast for Next Week	Weighted Moving Average Forecast for Next Week
June 16	$\frac{52 + 65 + 50}{3} = 55.7$ or 56	$[(0.5 \times 52) + (0.3 \times 65) + (0.2 \times 50)] = 55.5$ or 56
June 23	$\frac{56 + 52 + 65}{3} = 55.7$ or 58	$[(0.5 \times 56) + (0.3 \times 52) + (0.2 \times 65)] = 56.6$ or 57
June 30	$\frac{55 + 56 + 52}{3} = 54.3$ or 54	$[(0.5 \times 55) + (0.3 \times 56) + (0.2 \times 52)] = 54.7$ or 55
July 7	$\frac{60 + 55 + 56}{3} = 57.0$ or 57	$[(0.5 \times 60) + (0.3 \times 55) + (0.2 \times 56)] = 57.7$ or 58

Forecasts in each row are for the next week's demand. For example, the simple moving average and weighted moving average forecasts (both are 56 units) calculated after learning the demand on June 16 apply to June 23's demand forecast.

b. The mean absolute deviation is calculated as follows:

		SIMPLE MOVING AVERAGE		WEIGHTED MOVING AVERAGE	
Week	Actual Demand	Forecast for This Week	Absolute Errors $\lvert E_t \rvert$	Forecast for This Week	Absolute Errors $\lvert E_t \rvert$
June 23	56	56	$\lvert 56 - 56 \rvert = 0$	56	$\lvert 56 - 56 \rvert = 0$
June 30	55	58	$\lvert 55 - 58 \rvert = 3$	57	$\lvert 55 - 57 \rvert = 2$
July 7	60	54	$\lvert 60 - 54 \rvert = 6$	55	$\lvert 60 - 55 \rvert = 5$
			$\text{MAD} = \frac{0 + 3 + 6}{3} = 3.0$		$\text{MAD} = \frac{0 + 2 + 5}{3} = 2.3$

For this limited set of data, the weighted moving average method resulted in a slightly lower mean absolute deviation. However, final conclusions can be made only after analyzing much more data.

Discussion Questions

1. Kay and Michael Passe publish *What's Happening?*—a biweekly newspaper to publicize local events. *What's Happening?* has few subscribers; it typically is sold at checkout stands. Much of the revenue comes from advertisers of garage sales and supermarket specials. In an effort to reduce costs associated with printing too many papers or delivering them to the wrong location, Michael implemented a computerized system to collect sales data. Sales-counter scanners accurately record sales data for each location. Since the system was implemented, total sales volume has steadily declined. Selling advertising space and maintaining shelf space at supermarkets are getting more difficult.

Reduced revenue makes controlling costs all the more important. For each issue, Michael carefully makes a forecast based on sales data collected at each location. Then, he orders papers to be printed and distributed in quantities matching the forecast. Michael's forecast reflects a downward trend, which *is* present in the sales data. Now only a few papers are left over at only a few locations. Although the sales forecast accurately predicts the actual sales at most locations, *What's Happening?* is spiraling toward oblivion. Kay suspects that Michael is doing something wrong in preparing the forecast but can find no mathematical errors. Tell her what is happening.

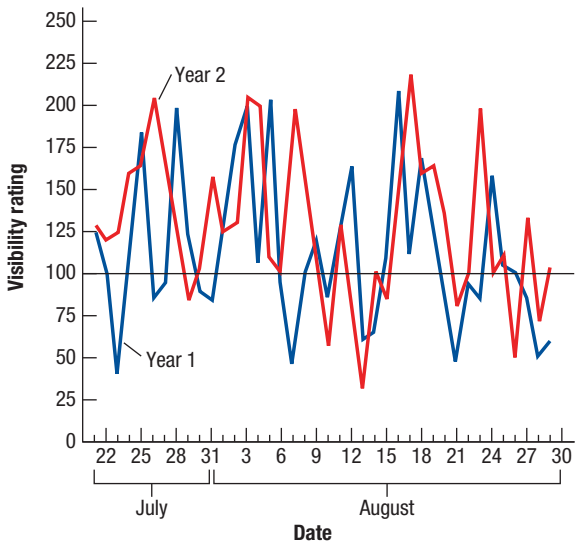
2. Figure 14.9 shows summer air visibility measurements for Denver, Colorado. The acceptable visibility standard is 100, with readings above 100 indicating clean air and good visibility, and readings below 100 indicating temperature inversions caused by forest fires, volcanic eruptions, or collisions with comets.

a. Is a trend evident in the data? Which time-series techniques might be appropriate for estimating the average of these data?

b. A medical center for asthma and respiratory diseases located in Denver has great demand for its services when air quality is poor. If you were in charge of developing a short-term (say, 3-day) forecast of visibility, which causal factor(s) would you analyze? In other words, which external factors hold the potential to significantly affect visibility in the *short term*?

c. Tourism, an important factor in Denver's economy, is affected by the city's image. Air quality, as measured by visibility, affects the city's image. If you were responsible for development of tourism, which causal factor(s) would you analyze to forecast visibility for the *medium term* (say, the next two summers)?

d. The federal government threatens to withhold several hundred million dollars in Department of Transportation funds unless Denver meets visibility standards within 8 years. How would you proceed to generate a *long-term* judgment forecast of technologies that will be available to improve visibility in the next 10 years?



▲ **FIGURE 14.9**
Summer Air Visibility Measurements

Problems

The OM Explorer and POM for Windows software is available to all students using the 10th edition of this textbook. Go to **www.pearsonglobaleditions.com/krajewski** to download these computer packages. If you purchased MyOMLab, you also have access to Active Models software and significant help in doing the following problems. Check with your instructor on how best to use these resources. In many cases, the instructor wants you to understand how to do the calculations by hand. At the least, the software provides a check on your calculations. When calculations are particularly complex and the goal is interpreting the results in making decisions, the software entirely replaces the manual calculations.

1. Forrest and Dan make boxes of chocolates for which the demand is uncertain. Forrest says, "That's life." On the other hand, Dan believes that some demand patterns exist that could be useful for planning the purchase of sugar, chocolate, and shrimp. Forrest insists on placing a surprise chocolate-covered shrimp in some boxes so that "You never know what you'll get." Quarterly demand (in boxes of chocolates) for the last 3 years follows:

Quarter	Year 1	Year 2	Year 3
1	3,000	3,300	3,502
2	1,700	2,100	2,448
3	900	1,500	1,768
4	4,400	5,100	5,882
Total	10,000	12,000	13,600

a. Use intuition and judgment to estimate quarterly demand for the fourth year.

b. If the expected sales for chocolates are 14,800 cases for year 4, use the multiplicative seasonal method to prepare a forecast for each quarter of the year. Are any of the quarterly forecasts different from what you thought you would get in part (a)?

2. Sales for the past 12 months at Dalworth Company are given here.

Month	Sales ($ millions)	Month	Sales ($ millions)
January	20	July	53
February	24	August	62
March	27	September	54
April	31	October	36
May	37	November	32
June	47	December	29

a. Use a three-month moving average to forecast the sales for the months May through December.

b. Use a four-month moving average to forecast the sales for the months May through December.

c. Compare the performance of the two methods by using the mean absolute deviation as the performance criterion. Which method would you recommend?

d. Compare the performance of the two methods by using the mean absolute percent error as the performance criterion. Which method would you recommend?

e. Compare the performance of the two methods by using the mean squared error as the performance criterion. Which method would you recommend?

3. Demand for oil changes at Garcia's Garage has been as follows:

Month	Number of Oil Changes
January	41
February	46
March	57
April	52
May	59
June	51
July	60
August	62

a. Use simple linear regression analysis to develop a forecasting model for monthly demand. In this application, the dependent variable, Y, is monthly demand and the independent variable, X, is the month. For January, let $X = 1$; for February, let $X = 2$; and so on.

b. Use the model to forecast demand for September, October, and November. Here, $X = 9$, 10, and 11, respectively.

4. The manager of a utility company in the Texas panhandle wants to develop quarterly forecasts of power loads for the next year. The power loads are seasonal, and the data on the quarterly loads in megawatts (MW) for the last 4 years are as follows:

Quarter	Year 1	Year 2	Year 3	Year 4
1	103.5	94.7	118.6	109.3
2	126.1	116.0	141.2	131.6
3	144.5	137.1	159.0	149.5
4	166.1	152.5	178.2	169.0

The manager estimates the total demand for the next year at 600 MW. Use the multiplicative seasonal method to develop the forecast for each quarter.

5. The following data are for calculator sales in units at an electronics store over the past 9 weeks:

Week	Sales	Week	Sales
1	46	6	58
2	49	7	62
3	43	8	56
4	50	9	63
5	53		

Use trend projection with regression to forecast sales for weeks 10–13. What are the error measures (CFE, MSE, σ, MAD, and MAPE) for this forecasting procedure? How about r^2?

6. Community Federal Bank in Dothan, Alabama, recently increased its fees to customers who use employees as tellers. Management is interested in whether its new fee policy has increased the number of customers now using its automatic teller machines to that point that more machines are required. The following table provides the number of automatic teller transactions by week. Use trend projection with regression to forecast usage for weeks 13–16.

Week	1	2	3	4	5	6	7	8	9	10	11	12
Transactions	716	721	833	639	689	736	779	711	723	835	829	667

7. The number of heart surgeries performed at Heartville General Hospital has increased steadily over the past several years. The hospital's administration is seeking the best method to forecast the demand for such surgeries in year 6. The data for the past 5 years are shown.

Year	Demand
1	45
2	50
3	52
4	56
5	58

The hospital's administration is considering the following forecasting methods. Begin error measurement in year 3, so all methods are compared for the same years.

i. Exponential smoothing, with $\alpha = 0.6$. Let the initial forecast for year 1 be 45, the same as the actual demand.

ii. Exponential smoothing, with $\alpha = 0.9$. Let the initial forecast for year 1 be 45, the same as the actual demand.

iii. Trend projection with regression.

iv. Two-year moving average.

v. Two-year weighted moving average, using weights 0.6 and 0.4, with more recent data given more weight.

vi. If MAD is the performance criterion chosen by the administration, which forecasting method should it choose?

vii. If MSE is the performance criterion chosen by the administration, which forecasting method should it choose?

viii. If MAPE is the performance criterion chosen by the administration, which forecasting method should it choose?

8. A convenience store recently started to carry a new brand of soft drink. Management is interested in estimating future sales volume to determine whether it should continue to carry the new brand or replace it with another brand. The table at the top of the next page provides the number of cans sold per week. Use both the trend projection with regression and the exponential smoothing (let $\alpha = 0.4$ with an initial forecast for week 1 of 617) methods to forecast demand for week 13. Compare these methods by using the mean absolute deviation and mean absolute percent error performance criteria. Does your analysis suggest that sales are trending and if so, by how much?

Week	1	2	3	4	5	6	7	8	9	10	11	12
Sales	617	617	648	739	659	623	742	704	724	715	668	740

9. The demand for Krispee Crunchies, a favorite breakfast cereal of people born in the 1940s, is experiencing a decline. The company wants to monitor demand for this product closely as it nears the end of its life cycle. The following table shows the actual sales history for January–October. Generate forecasts for November–December, using the trend projection by regression method. Looking at the accuracy of its forecasts over the history file, as well as the other statistics provided, how confident are you in these forecasts for November–December?

Month	Sales	Month	Sales
January	890,000	July	710,000
February	800,000	August	730,000
March	825,000	September	680,000
April	840,000	October	670,000
May	730,000	November	
June	780,000	December	

10. The owner of a computer store rents printers to some of her preferred customers. She is interested in arriving at a forecast of rentals so that she can order the correct quantities of supplies that go with the printers. Data for the last 10 weeks are shown here.

Week	Rentals	Week	Rentals
1	23	6	28
2	24	7	32
3	32	8	35
4	26	9	26
5	31	10	24

a. Prepare a forecast for weeks 6 through 10 by using a 5-week moving average. What is the forecast for week 11?

b. Calculate the mean absolute deviation as of the end of week 10.

11. The manager of Snyder's Garden Center must make the annual purchasing plans for rakes, gloves, and other gardening items. One of the items the company stocks is Fast-Grow, a liquid fertilizer. The sales of this item are seasonal, with peaks in the spring, summer, and fall months. Quarterly demand (in cases) for the past 2 years follows:

Quarter	Year 1	Year 2
1	40	60
2	350	440
3	290	320
4	210	280
Total	890	1,100

If the expected sales for Fast-Grow are 1,150 cases for year 3, use the multiplicative seasonal method to prepare a forecast for each quarter of the year.

12. Consider the sales data for Dalworth Company given in Problem 2.

a. Use a 3-month weighted moving average to forecast the sales for the months April through December. Use weights of (3/6), (2/6), and (1/6), giving more weight to more recent data.

b. Use exponential smoothing with $\alpha = 0.6$ to forecast the sales for the months April through December. Assume that the initial forecast for January was $22 million. Start error measurement in April.

c. Compare the performance of the two methods by using the mean absolute deviation as the performance criterion, with error measurement beginning in April. Which method would you recommend?

d. Compare the performance of the two methods by using the mean absolute percent error as the performance criterion, with error measurement beginning in April. Which method would you recommend?

e. Compare the performance of the two methods by using the mean squared error as the performance criterion, with error measurement beginning in April. Which method would you recommend?

13. Karl's Copiers sells and repairs photocopy machines. The manager needs weekly forecasts of service calls so that he can schedule service personnel. Use the actual demand in the first period for the forecast for the first week so error measurement begins in the second week. The manager uses exponential smoothing with $\alpha = 0.20$. Forecast the number of calls for week 6, which is next week.

Week	Actual Service Calls
1	24
2	32
3	36
4	23
5	25

14. At a hydrocarbon processing factory, process control involves periodic analysis of samples for a certain process quality parameter. The analytic procedure currently used is costly and time consuming. A faster and more economical alternative procedure has been proposed. However, the numbers for the quality parameter given by the alternative procedure are somewhat different from those given by the current procedure, not because of any inherent errors but because of changes in the nature of the chemical analysis.

Management believes that if the numbers from the new procedure can be used to forecast reliably the corresponding numbers from the current procedure, switching to the new procedure would be reasonable and cost effective. The following data were obtained for the quality parameter by analyzing samples using both procedures:

Current (Y)	Proposed (X)	Current (Y)	Proposed (X)
3.0	3.1	3.1	3.1
3.1	3.9	2.7	2.9
3.0	3.4	3.3	3.6
3.6	4.0	3.2	4.1
3.8	3.6	2.1	2.6
2.7	3.6	3.0	3.1
2.7	3.6	2.6	2.8

a. Use linear regression to find a relation to forecast Y, which is the quality parameter from the current procedure, using the values from the proposed procedure, X.

b. Is there a strong relationship between Y and X? Explain.

15. Ohio Swiss Milk Products manufactures and distributes ice cream in Ohio, Kentucky, and West Virginia. The company wants to expand operations by locating another plant in northern Ohio. The size of the new plant will be

a function of the expected demand for ice cream within the area served by the plant. A market survey is currently under way to determine that demand.

Ohio Swiss wants to estimate the relationship between the manufacturing cost per gallon and the number of gallons sold in a year to determine the demand for ice cream and, thus, the size of the new plant. The following data have been collected:

Plant	Cost per Thousand Gallons (*Y*)	Thousands of Gallons Sold (*X*)
1	$ 1,015	416.9
2	973	472.5
3	1,046	250.0
4	1,006	372.1
5	1,058	238.1
6	1,068	258.6
7	967	597.0
8	997	414.0
9	1,044	263.2
10	1,008	372.0
Total	$10,182	3,654.4

a. Develop a regression equation to forecast the cost per gallon as a function of the number of gallons produced.

b. What are the correlation coefficient and the coefficient of determination? Comment on your regression equation in light of these measures.

c. Suppose that the market survey indicates a demand of 325,000 gallons in the Bucyrus, Ohio, area. Estimate the manufacturing cost per gallon for a plant producing 325,000 gallons per year.

Advanced Problems

16. The Midwest Computer Company serves a large number of businesses in the Great Lakes region. The company sells supplies and replacements and performs service on all computers sold through seven sales offices. Many items are stocked, so close inventory control is necessary to assure customers of efficient service. Recently, business has been increasing, and management is concerned about stockouts. A forecasting method is needed to estimate requirements several months in advance so that adequate replenishment quantities can be purchased. An example of the sales growth experienced during the last 50 months is the growth in demand for item EP-37, a laser printer cartridge, shown in Table 14.3.

a. Develop a trend projection with regression solution using OM Explorer. Forecast demand for month 51.

b. A consultant to Midwest's management suggested that new office building leases would be a good leading indicator for company sales. The consultant quoted a recent university study finding that new office building leases precede office equipment and supply sales by 3 months. According to the study findings, leases in month 1 would affect sales in month 4, leases in month 2 would affect sales in month 5, and so on. Use POM for Windows' linear regression module to develop a forecasting model for sales, with leases as the independent variable. Forecast sales for month 51.

c. Which of the two models provides better forecasts? Explain.

17. A certain food item at P&Q Supermarkets has the demand pattern shown in the table at the bottom of the next page. There are 5 periods per cycle. Find the "best" forecast you can for month 25 and justify your methodology. If you wish to explore the Seasonal Forecasting method as one of the techniques tested, you will find that OM Explorer's *Seasonal Forecasting* Solver does not cover the case where there are 5 periods in a cycle (or seasons in a year). You must do some manual calculations or write an Excel spreadsheet on your own.

Period	Demand	Period	Demand
1	33	13	37
2	37	14	43
3	31	15	56
4	39	16	41
5	54	17	36
6	38	18	39
7	42	19	41
8	40	20	58
9	41	21	42
10	54	22	45
11	43	23	41
12	39	24	38

TABLE 14.3 | EP-37 SALES AND LEASE DATA

Month	EP-37 Sales	Leases	Month	EP-37 Sales	Leases
1	80	32	26	1,296	281
2	132	29	27	1,199	298
3	143	32	28	1,267	314
4	180	54	29	1,300	323
5	200	53	30	1,370	309
6	168	89	31	1,489	343
7	212	74	32	1,499	357
8	254	93	33	1,669	353
9	397	120	34	1,716	360
10	385	113	35	1,603	370
11	472	147	36	1,812	386
12	397	126	37	1,817	389
13	476	138	38	1,798	399
14	699	145	39	1,873	409
15	545	160	40	1,923	410
16	837	196	41	2,028	413
17	743	180	42	2,049	439
18	722	197	43	2,084	454
19	735	203	44	2,083	441
20	838	223	45	2,121	470
21	1,057	247	46	2,072	469
22	930	242	47	2,262	490
23	1,085	234	48	2,371	496
24	1,090	254	49	2,309	509
25	1,218	271	50	2,422	522

18. The director of a large public library must schedule employees to reshelf books and periodicals checked out of the library. The number of items checked out will determine the labor requirements. The following data reflect the number of items checked out of the library for the past 3 years:

Month	Year 1	Year 2	Year 3
January	1,847	2,045	1,986
February	2,669	2,321	2,564
March	2,467	2,419	2,635
April	2,432	2,088	2,150
May	2,464	2,667	2,201
June	2,378	2,122	2,663
July	2,217	2,206	2,055
August	2,445	1,869	1,678
September	1,894	2,441	1,845
October	1,922	2,291	2,065
November	2,431	2,364	2,147
December	2,274	2,189	2,451

The director needs a time-series method for forecasting the number of items to be checked out during the next month. Find the best simple moving average forecast you can. Decide what is meant by "best" and justify your decision.

19. Using the data in Problem 18, find the best exponential smoothing solution you can. Justify your choice.

20. Using the data in Problem 18, find the best trend projection with regression solution you can. Compare the performance of this method with those of the best moving average method (from Problem 18) and the exponential smoothing method (from Problem 19). Which of these three methods would you choose?

21. The materials handling manager of a manufacturing company is trying to forecast the cost of maintenance for the company's fleet of over-the-road tractors. The manager believes that the cost of maintaining the tractors increases with their age. The following data was collected:

Age (years)	Yearly Maintenance Cost ($)	Age (years)	Yearly Maintenance Cost ($)
5.5	819	5.0	1,394
6.5	1,249	1.5	363
5.5	1,233	1.5	382
4.0	695	8.0	964
5.0	923	7.0	1,573
5.0	881	2.0	1,178
5.0	1,090	2.0	666
6.0	1,722	2.0	749
6.5	1,187		

a. Use POM for Windows' least squares-linear regression module to develop a relationship to forecast the yearly maintenance cost based on the age of a tractor.

b. If a section has 15 four-year-old tractors, what is the forecast for the annual maintenance cost?

22. Franklin Tooling, Inc., manufactures specialty tooling for firms in the paper-making industry. All of their products are engineer-to-order and so the company never knows exactly what components to purchase for a tool until a customer places an order. However, the company believes that weekly demand for a few components is fairly stable. Component 135.AG is one such item. The last 26 weeks of historical use of component 135.AG is recorded below.

Week	Demand	Week	Demand
1	137	14	131
2	136	15	132
3	143	16	124
4	136	17	121
5	141	18	127
6	128	19	118
7	149	20	120
8	136	21	115
9	134	22	106
10	142	23	120
11	125	24	113
12	134	25	121
13	118	26	119

Use OM Explorer's *Time Series Forecasting* Solver to evaluate the following forecasting methods. Start error measurement in the fifth week, so all methods are evaluated over the same time interval. Use the default settings for initial forecasts.

i. Naïve (1-Period Moving Average)

ii. 3-Period Moving Average

iii. Exponential Smoothing, with $\alpha = .28$

iv. Trend Projection with Regression

v. Which forecasting method should management use, if the performance criterion it chooses is:

- CFE?
- MSE?
- MAD?
- MAPE?

23. Create an Excel spreadsheet on your own that can create combination forecasts for Problem 22. Create a combination forecast using all four techniques from problem 22. Give each technique an equal weight. Create a second combination forecast by using the three techniques that seem best based on MAD. Give equal weight to each technique. Finally, create a third forecast by equally weighting the two best techniques. Calculate CFE, MAD, MSE, and MAPE for the combination forecast. Are these forecasts better or worse than the forecasting techniques identified in Problem 22?

24. The data for the visibility chart in Discussion Question 1 are shown in Table 14.4. The visibility standard is set at 100. Readings below 100 indicate that air pollution has reduced visibility, and readings above 100 indicate that the air is clearer.

a. Use several methods to generate a visibility forecast for August 31 of the second year. Which method seems to produce the best forecast?

b. Use several methods to forecast the visibility index for the summer of the third year. Which method seems to produce the best forecast? Support your choice.

25. Tom Glass forecasts electrical demand for the Flatlands Public Power District (FPPD). The FPPD wants to take its Comstock power plant out of service for maintenance when demand is expected to be low. After shutdown, performing maintenance and getting the plant back on line takes two weeks. The utility has enough other generating capacity to satisfy 1,550 megawatts (MW) of demand while Comstock is out of service. Table 14.5 at the end of the Advanced Problems shows weekly peak demands (in MW) for the past several autumns. When next fall should the Comstock plant be scheduled for maintenance?

26. A manufacturing firm has developed a skills test, the scores from which can be used to predict workers' production rating factors. Data on the test scores of various workers and their subsequent production ratings are shown.

TABLE 14.4 | VISIBILITY DATA

Date	Year 1	Year 2	Date	Year 1	Year 2	Date	Year 1	Year 2
July 22	125	130	Aug 5	105	200	Aug 19	170	160
23	100	120	6	205	110	20	125	165
24	40	125	7	90	100	21	85	135
25	100	160	8	45	200	22	45	80
26	185	165	9	100	160	23	95	100
27	85	205	10	120	100	24	85	200
28	95	165	11	85	55	25	160	100
29	200	125	12	125	130	26	105	110
30	125	85	13	165	75	27	100	50
31	90	105	14	60	30	28	95	135
Aug 1	85	160	15	65	100	29	50	70
2	135	125	16	110	85	30	60	105
3	175	130	17	210	150			
4	200	205	18	110	220			

TABLE 14.5 | WEEKLY PEAK POWER DEMANDS

	AUGUST			SEPTEMBER				OCTOBER				NOVEMBER	
Year	1	2	3	4	5	6	7	8	9	10	11	12	13
1	2,050	1,925	1,825	1,525	1,050	1,300	1,200	1,175	1,350	1,525	1,725	1,575	1,925
2	2,000	2,075	2,225	1,800	1,175	1,050	1,250	1,025	1,300	1,425	1,625	1,950	1,950
3	1,950	1,800	2,150	1,725	1,575	1,275	1,325	1,100	1,500	1,550	1,375	1,825	2,000
4	2,100	2,400	1,975	1,675	1,350	1,525	1,500	1,150	1,350	1,225	1,225	1,475	1,850
5	2,275	2,300	2,150	1,525	1,350	1,475	1,475	1,175	1,375	1,400	1,425	1,550	1,900

Worker	Test Score	Production Rating	Worker	Test Score	Production Rating
A	58	50	K	68	64
B	41	56	L	78	63
C	96	94	M	70	51
D	89	84	N	34	33
E	91	89	O	60	56
F	69	77	P	27	32
G	51	54	Q	81	84
H	53	53	R	37	39
I	44	39	S	51	65
J	72	81	T	42	37

a. Using POM for Windows' least squares-linear regression module, develop a relationship to forecast production ratings from test scores.

b. If a worker's test score was 80, what would be your forecast of the worker's production rating?

c. Comment on the strength of the relationship between the test scores and production ratings.

27. A manufacturing firm seeks to develop a better forecast for an important product, and believes that there is a trend to the data. OM Explorer's *Trend Projection with Regression* Solver has been set up with the 47 demands in the history file. Note the "Load Problem 26 Data" button in the *Trend Projection with Regression* Solver that when clicked will automatically input the demand data. Otherwise, you can enter the demand data directly into the Inputs sheet.

Yr	1	2	3	4
Jan	4507	4589	4084	4535
Feb	4400	4688	4158	4477
Mar	4099	4566	4174	4601
Apr	4064	4485	4225	4648
May	4002	4385	4324	4860
Jun	3963	4377	4220	4998
Jul	4037	4309	4267	5003
Aug	4162	4276	4187	4960
Sep	4312	4280	4239	4943
Oct	4395	4144	4352	5052
Nov	4540	4219	4331	5107
Dec	4471	4052	4371	

a. What is your forecast for December of Year 4, making period 1 as the starting period for the regression?

b. The actual demand for period 48 was just learned to be 5,100. Add this demand to the Inputs file and change the starting period for the regression to period 2 so that the number of periods in the regression remains unchanged. How much or little does the forecast for period 49 change from the one for period 48? The error measures? Are you surprised?

c. Now change the time when the regression starts to period 25 and repeat the process. What differences do you note now? What forecast will you make for period 49?

28. Cannister, Inc., specializes in the manufacture of plastic containers. The data on the monthly sales of 10-ounce shampoo bottles for the past 5 years are as follows:

Year	1	2	3	4	5
January	742	741	896	951	1,030
February	697	700	793	861	1,032
March	776	774	885	938	1,126
April	898	932	1,055	1,109	1,285
May	1,030	1,099	1,204	1,274	1,468
June	1,107	1,223	1,326	1,422	1,637
July	1,165	1,290	1,303	1,486	1,611
August	1,216	1,349	1,436	1,555	1,608
September	1,208	1,341	1,473	1,604	1,528
October	1,131	1,296	1,453	1,600	1,420
November	971	1,066	1,170	1,403	1,119
December	783	901	1,023	1,209	1,013

a. Using the multiplicative seasonal method, calculate the monthly seasonal indices.

b. Develop a simple linear regression equation to forecast annual sales. For this regression, the dependent variable, Y, is the demand in each year and the independent variable, X, is the index for the year (i.e., $X = 1$ for year 1, $X = 2$ for year 2, and so on until $X = 5$ for year 5).

c. Forecast the annual sales for year 6 by using the regression model you developed in part (b).

d. Prepare the seasonal forecast for each month by using the monthly seasonal indices calculated in part (a).

VIDEO CASE Forecasting and Supply Chain Management at Deckers Outdoor Corporation

Deckers Outdoor Corporation's footwear products are among some of the most well-known brands in the world. From UGG sheepskin boots and Teva sport sandals to Simple shoes, Deckers flip-flops, and Tsubo footwear, Deckers is committed to building niche footwear brands into global brands with market leadership positions. Net sales for fiscal year 2007 were close to $449 million. In addition to traditional retail store outlets for Deckers' footwear styles, the company maintains an active and growing "direct to consumer" e-commerce business. Since most retail stores cannot carry every style in every color and size, the company offers the full line for each of its brands directly to consumers through the brands' individual Web sites. Online sales at its virtual store are handled by its e-commerce group. Customers who want a pair of shoes not available at the retail store can always buy from the virtual store.

Founded in 1973, the company manufactured a single line of sandals in a small factory in Southern California. The challenges of managing the raw materials and finished goods inventories were small compared to today's global sourcing and sales challenges for the company's various brands. Today, each brand has its own development team and brand managers who generate, develop, and test-market the seasonal styles that appear on the shelves of retailers such as Nordstrom, Lord & Taylor, REI, the Walking Company, and the company's own UGG brand retail stores in the United States and Japan.

At Deckers, forecasting is the starting point for inventory management, sales and operations planning, resource planning, and scheduling—in short, managing its supply chain. It carries a considerable amount of seasonal stock. Shoes with seasonal demand that are left over at the end of their season must be sold at heavily discounted prices. Its products fall into three categories: (1) carry-over items that were sold in prior years, (2) new items that look similar to past models, and (3) completely new designs that are fashionable with no past history.



Pearson

Twice a year, the brand development teams work on the fall and spring product lines. They come up with new designs about one year in advance of each season. Each brand (UGG, Teva, Simple, Tsubo, and Deckers) contains numerous products, called stock keeping units (SKUs). The materials for new designs are selected and tested in prototypes. Approved designs are put into the seasonal line-up. Forecasts must be made at both the SKU and aggregate levels months before the season begins. "Bottoms-up" forecasts for each SKU begin by analyzing any available history files of past demand. Judgment forecasts are also important inputs, particularly for the second and third categories of shoes that are not carry-overs. For example, Char Nicanor-Kimball is an expert in spotting trends in shoe sales and makes forecasts for the virtual store. For new designs, historical sales on similar items are used to make a best guess on demand for those items. This process is facilitated by a forecasting and inventory system on the company's Intranet. At the same time, the sales teams for each brand call on their retail accounts and secure customer orders of approved designs for the coming season. Then, the virtual store forecasts are merged with orders from the retail store orders to get the total seasonal demand forecasted by SKU. Next, the SKU forecasts are "rolled up" by category and "top down" forecasts are also made.

These forecasts then go to top management where some adjustments may be made to account for financial market conditions, consumer credit, weather, demographic factors, and customer confidence. The impact of public relations and advertising must also be considered.

Actually, forecasting continues on throughout the year on a daily and weekly basis to "get a handle" on demand. Comparing actual demand with what was forecasted for different parts of the season also helps the forecasters make better forecasts for the future and better control inventories.

Based on initial demand forecasts, the company must begin sourcing the materials needed to produce the footwear. The company makes most of its products in China and sources many of the raw materials there as well. For UGG products sheepskin sourcing occurs in Australia with top grade producers, but the rawhide tanning still takes places in China. With potential suppliers identified and assurance from internal engineering that the footwear can be successfully made, the engineering and material data are handed over to the manufacturing department to determine how best to make the footwear in mass quantities. At this point, Deckers places a seasonal "buy" with its suppliers.

The orders for each SKU are fed into the manufacturing schedules at the Chinese factories. All the SKUs for a given brand are manufactured at the same factory. While Deckers agents negotiate the raw materials contracts early in the development process, the factories only place the orders for the raw materials when the company sends in the actual orders for the finished goods. No footwear is made by the factories until orders are received.

At the factories, finished goods footwear is inspected and packaged for the month-long ocean voyage from Hong Kong to ports in the United States. Deckers ships fifty containers a week from its Chinese manufacturing sources, each holding approximately 5,000 pairs of shoes. Ownership of the finished goods transfers from the factories to Deckers in Hong Kong.

When the shipping containers arrive in the United States, the footwear is transferred to Deckers' distribution centers in Southern California. Teva products are warehoused in Ventura, California; all other products are handled by the company's state-of-the-art facility in Camarillo, California. Typically, Deckers brings product into the distribution centers two to three months in advance of expected needs so that the production at the suppliers' factories and the labor activities at the distribution centers are leveled. There are definitive spikes in the demand for footwear, with Teva spiking in Quarter 1 and UGG spiking in Quarter 4. The leveling approach works to keep costs low in the supply chain. However, it also means that Deckers must maintain sizeable inventories. Most shipments from suppliers come in to the distribution centers and are stored in inventory for one to two months awaiting a customer order. By the time the footwear is stocked in the distribution center, the company knows which retail customers will be getting the various products, based on the orders booked months earlier. Then, according to delivery schedules negotiated with the customers, the company begins filling orders and shipping products to retail locations. The warehouse tracks incoming shipments, goods placed on the shelves for customers, and outgoing orders. The inventory system helps manage the customer order filling process.

Because the booked orders are a relatively large proportion of the total orders from retailers, and the number of unanticipated orders is very small, only small safety stocks are needed to service the retailers. Occasionally, the purchase order from Deckers to one of its suppliers matches the sales order from the customer. In such a case, Deckers uses a "cross-dock" system. When the shipment is received at the distribution center, it is immediately checked in and loaded on another truck for delivery to customers. Cross docking reduces the need to store vast quantities of product for long periods of time and cuts down on warehousing expenses for Deckers. The company has been successful in turning its inventory over about four times a year, which is in line with footwear industry standards.

The online sales traffic is all managed centrally. In fact, for ordering and inventory management purposes, the online side of the business is treated just like another major retail store account. As forecasted seasonal orders are generated by each brand's sales team, a manufacturing order for the online business is placed by the e-commerce sales team at the same time. However, unlike the retail outlets that take delivery of products on a regular schedule, the inventory pledged to the online business is held in the distribution center until a Web site order is received. Only then is it shipped directly to the consumer who placed the online order. If actual demand exceeds expected demand, Char Nicanor-Kimball checks if more inventory can be secured from other customer orders that have scaled back.

The forecasting and supply chain management challenges now facing Deckers are two-fold. First, the company plans to grow the brands that have enjoyed seasonal sales activity into year-round footwear options for consumers by expanding the number of SKUs for those brands. For example, most sales for UGG footwear occur in the fall/winter season. Sales for Teva historically have been in the spring and summer. Product managers are now working to develop styles that will allow the brands to cross over the seasons. Second, the company plans to expand internationally, and will have retail outlets in Europe, China, and other Asian locations in the very near future. Company managers are well aware of the challenges and opportunities such global growth will bring, and are taking steps now to assure that the entire supply chain is prepared to forecast and handle the demand when the time comes.

QUESTIONS

1. How much does the forecasting process at Deckers correspond with the "typical forecasting process" described at the end of this chapter?
2. Based on what you see in the video, what kinds of information technology are used to make forecasts, maintain accurate inventory records, and project future inventory levels?
3. What factors make forecasting at Deckers particularly challenging? How can forecasts be made for seasonal, fashionable products for which there is no history file? What are the costs of over-forecasting demand for such items? Under-forecasting?
4. How does the concept of *postponement* get implemented at Deckers by having online sales and positioning inventory at the DCs for every model, color, and size?
5. Where in the supply chain are cycle, pipeline, safety stock, and anticipation inventories being created?
6. What are the benefits of leveling aggregate demand by having a portfolio of SKUs that create 365-day demand?
7. Deckers plans to expand internationally, thereby increasing the volume of shoes it must manage in the supply chain and the pattern of material flows. What implications does this strategy have on forecasting, order quantities, logistics, and relationships with its suppliers and customers?

CASE Yankee Fork and Hoe Company

The Yankee Fork and Hoe Company is a leading producer of garden tools ranging from wheelbarrows, mortar pans, and hand trucks to shovels, rakes, and trowels. The tools are sold in four different product lines ranging from the top-of-the-line Hercules products, which are rugged tools for the toughest jobs, to the Garden Helper products, which are economy tools for the occasional user. The market for garden tools is extremely competitive because of the simple design of the products and the large number of competing producers. In addition, more people are using power tools, such as lawn edgers, hedge trimmers, and thatchers, reducing demand for their manual counterparts. These factors compel Yankee to maintain low prices while retaining high quality and dependable delivery.

Garden tools represent a mature industry. Unless new manual products can be developed or a sudden resurgence occurs in home gardening, the prospects for large increases in sales are not bright. Keeping ahead of the competition is a constant battle. No one knows this better than Alan Roberts, president of Yankee.

The types of tools sold today are, by and large, the same ones sold 30 years ago. The only way to generate new sales and retain old customers is to provide superior customer service and produce a product with high customer value. This approach puts pressure on the manufacturing system, which has been having difficulties lately. Recently, Roberts has been receiving calls from long-time customers, such as Sears and True Value Hardware Stores, complaining about late shipments. These customers advertise promotions for garden tools and require on-time delivery.

Roberts knows that losing customers like Sears and True Value would be disastrous. He decides to ask consultant Sharon Place to look into the matter and report to him in one week. Roberts suggests that she focus on the bow rake as a case in point because it is a high-volume product and has been a major source of customer complaints of late.

Planning Bow Rake Production

A bow rake consists of a head with 12 teeth spaced 1 inch apart, a hardwood handle, a bow that attaches the head to the handle, and a metal ferrule that reinforces the area where the bow inserts into the handle. The bow is a metal strip that is welded to the ends of the rake head and bent in the middle to form a flat tab for insertion into the handle. The rake is about 64 inches long.

Place decides to find out how Yankee plans bow rake production. She goes straight to Phil Stanton, who gives the following account:

> Planning is informal around here. To begin, marketing determines the forecast for bow rakes by month for the next year. Then they pass it along to me. Quite frankly, the forecasts are usually inflated—must be their big egos over there. I have to be careful because we enter into long-term purchasing agreements for steel, and having it just sitting around is expensive. So I usually reduce the forecast by 10 percent or so. I use the modified forecast to generate a monthly final-assembly schedule, which determines what I need to have from the forging and woodworking areas. The system works well if the forecasts are good. But when marketing comes to me and says they are behind on customer orders, as they often do near the end of the year, it wreaks havoc with the schedules. Forging gets hit the hardest. For example, the presses that stamp the rake heads from blanks of steel can handle only 7,000 heads per day, and the bow rolling machine can do only 5,000 per day. Both operations are also required for many other products.

Because the marketing department provides crucial information to Stanton, Place decides to see the marketing manager, Ron Adams. Adams explains how he arrives at the bow rake forecasts.

> Things do not change much from year to year. Sure, sometimes we put on a sales promotion of some kind, but we try to give Phil enough warning before the demand kicks in—usually a month or so. I meet with several managers from the various sales regions to go over shipping data from last year and discuss anticipated promotions, changes in the economy, and shortages we experienced last year. Based on these meetings, I generate a monthly forecast for the next year. Even though we take a lot of time getting the forecast, it never seems to help us avoid customer problems.

The Problem

Place ponders the comments from Stanton and Adams. She understands Stanton's concerns about costs and keeping inventory low and Adams's concern about having enough rakes on hand to make timely shipments. Both are also somewhat concerned about capacity. Yet she decides to check actual customer demand for the bow rake over the past 4 years (in Table 14.6) before making her final report to Roberts.

QUESTIONS

1. Comment on the forecasting system being used by Yankee. Suggest changes or improvements that you believe are justified.
2. Develop your own forecast for bow rakes for each month of the next year (year 5). Justify your forecast and the method you used.

TABLE 14.6 | FOUR-YEAR DEMAND HISTORY FOR THE BOW RAKE

	DEMAND			
Month	**Year 1**	**Year 2**	**Year 3**	**Year 4**
1	55,220	39,875	32,180	62,377
2	57,350	64,128	38,600	66,501
3	15,445	47,653	25,020	31,404
4	27,776	43,050	51,300	36,504
5	21,408	39,359	31,790	16,888
6	17,118	10,317	32,100	18,909
7	18,028	45,194	59,832	35,500
8	19,883	46,530	30,740	51,250
9	15,796	22,105	47,800	34,443
10	53,665	41,350	73,890	68,088
11	83,269	46,024	60,202	68,175
12	72,991	41,856	55,200	61,100

Note: The demand figures shown in the table are the number of units promised for delivery each month. Actual delivery quantities differed because of capacity or shortages of materials.

EXPERIENTIAL LEARNING 14.1 Forecasting with Holdout Sample

A company's history file, as shown in the following table, gives monthly sales in thousands of dollars "rolled up" into aggregated totals for one of its major product lines.

Your team should use the *Time Series Forecasting* Solver to make forecasts into the future. Note the "Load EL1 Data" button in this solver when clicked will automatically input the demand data. Otherwise, you can enter the demand data directly into the Inputs sheet. Seek out which models you wish to use in making in-class forecasts of monthly sales for the last two months of year 8 and several months into year 9. Perhaps you might want to know the forecasts of all of them, or alternately focus on just two or three of them. If one of the models is the combination forecast, you must decide on the weights to give the models going into its forecast. The weights should add up to 1.0.

Bring to class a one-page document that

- characterizes the monthly sales of the product line in terms of its forecastability.
- identifies the relative importance of four demand patterns: horizontal, trend, seasonal, and cyclical.
- identifies the forecasting models that you will use to make the forecasts for the last of year 8, and future months into year 9, and the extent that judgmental adjustments might be used during the holdout sample exercise. Explain why you made this selection, given that MAD will be used as your error measure.
- makes the November forecast for year 8.

At the start of the in-class portion of the experiential exercise, hand in your one-page document and open the final *Time Series Forecasting* Solver file that you used in modeling the history file. Do not change any of the final parameters chosen for your various forecasting models using the history file (n for moving average, weights for weighted moving average, α for exponential smoothing, and weights for combination).

To start the Holdout Sample session, click on the Worksheet tab and set the time when error analysis begins to be period 95 (November, Year 8). In doing so, error analysis will be tracked only for the holdout periods. Now click on the "Holdout Sample" tab to begin the session. You will initially be presented with the November forecasts for all of the techniques used during your analysis of the history file (including the combination model if you used it). Your next step is to input your team's forecast for November. It can be the forecast from any of the techniques shown, or one of your own if you believe that judgmental adjustment is appropriate. You have no contextual information, but may observe that one model has been performing particularly well in the last few months. You team might have different opinions, but you must reach a consensus. Your instructor will then provide November's actual sales from the holdout sample. After you input that additional information, forecast errors are computed for each model and for your team's November forecast. In addition, computer forecasts (naïve, moving average, weighted moving average, exponential smoothing, trend projection, and combination) are posted for December.

Begin December by inputting your team's forecast. The instructor then provides December's actual sales, and so forth. Continue this process until all errors are calculated for the last period in the holdout sample, which will be announced by your instructor. At the end of this exercise, create a second one-page document that reports your forecasts for the holdout sample, the corresponding average MAD, and CFE whether (and how) you modified your forecasting process as the exercise progressed, and what you learned from this exercise. You will need to write an Excel spreadsheet to calculate the MAD and CFE statistics for the holdout sample. Its output can be attached to your second one-page document. Submit your report to your instructor at the end of the class session.

Your grade on this exercise will be based on (1) the insights provided in the two documents (50 percent of grade) (2) the average MAD for the history file (25 percent of grade), and (3) the average MAD for the holdout sample (25 percent of grade).

Yr	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1	3,255	3,420	3,482	3,740	3,713	3,785	3,817	3,900	3,878	3,949	4,004	4,035
2	3,892	3,730	4,115	4,054	4,184	4,321	4,307	4,481	4,411	4,443	4,395	4,403
3	4,507	4,400	4,099	4,064	4,002	3,963	4,037	4,162	4,312	4,395	4,540	4,471
4	4,589	4,688	4,566	4,485	4,385	4,377	4,309	4,276	4,280	4,144	4,219	4,052
5	4,084	4,158	4,174	4,225	4,324	4,220	4,267	4,187	4,239	4,352	4,331	4,371
6	4,535	4,477	4,601	4,648	4,860	4,998	5,003	4,960	4,943	5,052	5,107	5,100
7	5,303	5,550	5,348	5,391	5,519	5,602	5,557	5,608	5,663	5,497	5,719	5,679
8	5,688	5,604	5,703	5,899	5,816	5,745	5,921	5,900	5,911	5,987		

Sources: This experiential exercise was adapted from an in-class exercise prepared by Dr. Richard J. Penlesky, Carroll University, as a basis for classroom discussion. By permission of Richard J. Penlesky.

EXPERIENTIAL LEARNING 14.2 Forecasting a Vital Energy Statistic

The following time series data captures the weekly average of East Coast crude oil imports in thousands of barrels per day.

QUARTER 2 2010		QUARTER 3 2010		QUARTER 4 2010		QUARTER 1 2011	
Time Period	Data	Time Period	Data	Time Period	Data	Time Period	Data
Apr 02, 2010	1,160	Jul 02, 2010	1,116	Oct 01, 2010	1,073	Dec 31, 2010	994
Apr 09, 2010	779	Jul 09, 2010	1,328	Oct 08, 2010	857	Jan 07, 2011	1,307
Apr 16, 2010	1,134	Jul 16, 2010	1,183	Oct 15, 2010	1,197	Jan 14, 2011	997
Apr 23, 2010	1,275	Jul 23, 2010	1,219	Oct 22, 2010	718	Jan 21, 2011	1,082
Apr 30, 2010	1,355	Jul 30, 2010	1,132	Oct 29, 2010	817	Jan 28, 2011	887
May 07, 2010	1,513	Aug 06, 2010	1,094	Nov 05, 2010	946	Feb 04, 2011	1,067
May 14, 2010	1,394	Aug 13, 2010	1,040	Nov 12, 2010	725	Feb 11, 2011	890
May 21, 2010	1,097	Aug 20, 2010	1,053	Nov 19, 2010	748	Feb 18, 2011	865
May 28, 2010	1,206	Aug 27, 2010	1,232	Nov 26, 2010	1,031	Feb 25, 2011	858
Jun 04, 2010	1,264	Sep 03, 2010	1,073	Dec 03, 2010	1,061	Mar 04, 2011	814
Jun 11, 2010	1,153	Sep 10, 2010	1,329	Dec 10, 2010	1,074	Mar 11, 2011	871
Jun 18, 2010	1,424	Sep 17, 2010	1,096	Dec 17, 2010	941	Mar 18, 2011	1,255
Jun 25, 2010	1,274	Sep 24, 2010	1,125	Dec 24, 2010	994	Mar 25, 2011	980

Your instructor has a "holdout" sample representing the values for April 1, 2011 and beyond. Your task is to use the POM for Windows *Time Series Forecasting* module and the history file to project this statistic into the future. If you have MyOMLab, the demand data is available in the *Exercise 2* Excel file. It can be pasted into the Data Table of the *Time Series Forecasting* module. Otherwise, you can enter the demand data directly into the Data Table. Prior to your next class meeting:

a. Use the POM for Windows *Time Series Forecasting* module to locate the best naïve, moving average, weighted moving average, and trend

projection with regression models that you think will most accurately forecast demand during the holdout sample. *Begin your error calculations with period 5 (April 30, 2010).*

b. Create an Excel spreadsheet that begins with inputs of the four forecasts from the *Time Series Forecasting* module. Its purpose is to develop a combination forecast that will serve as your team's forecasts for each period. Assign a weight to each forecast model (the sum of all four forecast weights for one period should equal 1.0) and develop a "combination forecast" by multiplying each forecast by its weight. Keep the weights constant for the whole history file as you search for the best set of weights. If you do not like a particular model, give it a weight of 0. Calculate appropriate forecast error measures for your combination forecast in your Excel spreadsheet.

c. Create a management report that shows your period-by-period forecasts and their overall historical CFE and MAPE performance for each model and your combination forecast.

In-Class Exercise–Part 1

a. Input into your Excel spreadsheet the forecasts from the POM for Windows *Time Series Forecasting* module to get the combination forecast for the first period (the week of April 1, 2011) in the holdout sample. The combination forecast is considered your team's forecast.

b. Enter the actual data announced by your instructor, and have Excel compute appropriate forecast error measures for your four models and the combination forecast. Decide on any revisions of weights for the combination forecast.

c. Update the POM for Windows *Time Series Forecasting* module with the actual demand for the new period and get the new forecasts.

In-Class Exercise–Part 2

a. Input the forecasts from the POM for Windows *Time Series Forecasting* module into your Excel spreadsheet to get the final combination forecast for the next period (the week of April 8, 2011). At this point, you may change this period's weights on each forecasting technique going into the combination forecast. You have no contextual information, but may observe that one model has been performing particularly well in the last few periods. Your team might have different opinions, but you must reach a consensus.

b. Enter the actual data announced by your instructor, with Excel computing appropriate forecast error measures for your four models and the combination forecast.

c. Update the POM for Windows *Time Series Forecasting* module with the actual demand for the new period and get the new forecasts.

In-Class Exercise–Parts 3 and beyond

Continue in the fashion of Parts 1 and 2 to produce forecasts as directed by your instructor. At the end of the exercise, create a second management report that shows for the holdout sample your period-by-period forecasts, their individual forecast errors and percent deviations for each model and your combination forecast. Explain your logic regarding any changes made to your combination forecast weights over the holdout period.

Source: This experiential exercise was prepared as an in-class exercise prepared by Dr. John Jensen, University of South Carolina, as a basis for classroom discussion. By permission of John B. Jensen.

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